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More Than Money: Job Quality, Employment, and Occupational Change through the Pandemic

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Summary:
We study how the quality of an occupation, measured by the autonomy it gives workers and by how exposed its work is, bore on two decisions through the COVID-19 pandemic. The first is whether a worker employed on one date was also employed a year later, and the second is whether a worker employed on both dates changed occupation. We build autonomy and safety scores for 474 occupations from the Occupational Requirements Survey and O*NET and attach them to matched Current Population Survey records for 2018 to 2025. We find that workers in occupations with more autonomy were more likely to be employed a year later during the pandemic, but not before or after it. Whether a worker changed occupation depended on job quality, and in particular on the autonomy of the occupations that workers leaving the same occupation moved into and on the safety of the occupation the worker held. We find no change across the periods in the effect of job quality on occupation change. Workers who changed occupation gained slightly more autonomy after March 2020 than before, and slightly more safety after 2022, although these gains are small.

JEL classification: J62, J63, J81, J28, J16, I18

Key words: job quality, occupational mobility, working conditions, autonomy, COVID-19, Current Population Survey

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I. Introduction

When describing the virtues of various occupations the focus is typically on wages and whether a job is high- or low-paying. However, there are many characteristics other than wage on which occupations/jobs differ, and individuals have preferences about those characteristics. For example, working in an office at a desk would be torture to one individual, but preferred by another. Additionally, some shared events, such as the COVID-19 pandemic, may influence a collective preference for certain characteristics, such as wanting to avoid occupations that involve interactions with the public or a newfound appreciation for telework (Barrero, Bloom, and Davis 2021; Barrero, Bloom, Davis, Meyer, and Mihaylov 2022). One's own experience through the pandemic may have altered individual preferences. For example, restaurant workers interviewed during the pandemic described fear of exposure under inadequate safety policies, job insecurity, and inconsistent pay and hours, which left many uncertain whether they would return to the industry (Lippert et al. 2021). A survey of more than 2,800 food service workers conducted between October 2020 and May 2021 found that tipped restaurant workers reported increased health risks and harassment during the pandemic and that 53 percent were considering leaving their jobs, most often because of low wages and tips and concerns about COVID-19 safety (One Fair Wage and UC Berkeley Food Labor Research Center 2021).

The purpose of this article is to explore whether the weight workers placed on these characteristics changed in the years surrounding the pandemic as reflected in how the distribution of workers across occupational characteristics changed over this period and whether any such changes can be tied to individual workers' employment and occupational choice decisions. To that end, we focus on the characteristics of the occupations whose desirability may have been most impacted by the pandemic. One possible change for workers could be in

preferences related to potential exposure, including the ability to telework, interact with others, and work around crowds. Our scores capture telework only indirectly, since it carries little weight in the safety score, although the autonomy score is highly correlated with it. Additionally, after the freedom granted to many workers during the periods of forced remote work, preferences related to autonomy may have changed. These are measured by the ability to determine one's own pace, to take breaks, and to sit or stand as needed. We explore how these characteristics affect year-over-year employment and occupation changes, conditioning on state pandemic policies, such as mask and vaccine mandates as well as business and school closures.

In a tighter labor market, when workers enjoy greater bargaining power, workers may be able to move to occupations with more desirable characteristics. Additionally, an experience such as the pandemic may result in a reassessment of priorities, and we may observe workers shifting to jobs that allow telework opportunities or reduce interaction with the public. Because the occupation scores are fixed, a change across the periods in how they bear on these decisions could reflect a change in how workers weighed the characteristics, but it could also reflect changes in the demand for those occupations or in who holds them.

We find that different factors affected the two outcomes studied here: (1) how likely a worker was to be observed employed from one year to the next, and (2) whether a worker changed occupations over the time period. The autonomy of an occupation had no significant effect on whether a worker was employed at both dates before March 2020 or after 2022, but during the pandemic workers in more autonomous occupations were more likely to be employed a year later, an effect that likely overlaps the ability to work from home. Our results suggest that whether a worker changed occupation depended on the autonomy of the occupations that workers leaving the same occupation moved into and on the safety of the occupation the worker

held. Workers in occupations with more autonomy were also somewhat more likely to change occupation, but this effect is smaller and less precisely estimated. We find no change across the periods in the effect of job quality on occupation change. Workers who changed occupation gained slightly more autonomy after March 2020 than before, and slightly more safety after 2022, although these gains are small.[1] The average autonomy and safety of employed workers rose over the period, with a step in autonomy in 2020 and one in safety in 2021 (Online Appendix Figure OA7). The step in autonomy is consistent with a change in composition, since the pandemic job losses fell most on low-autonomy occupations, rather than with a reassessment set off by the pandemic. Taken together, we find no evidence that the pandemic changed the weight workers placed on the quality of their jobs when deciding whether to change occupation.

II. Background

A. Valuing the Non-Wage Features of a Job

The premise that workers trade pay against the conditions of the work is not new. Rosen (1986) sets out the theory of equalizing differences, in which an observed wage is the price of a bundle of attributes rather than of labor alone, so that jobs with worse conditions must pay more to attract the same worker. The difficulty has always been empirical as conditions are hard to observe, workers sort on them, and the cross-sectional wage differences that should reveal the trade-offs are confounded by everything else that differs across jobs.

Maestas, Mullen, Powell, von Wachter, and Wenger (2023) address this with stated-preference experiments validated against actual job choices. They find that working conditions vary substantially across jobs, that they play a significant role in which job a worker takes, and that they are a central component of compensation rather than a residual; accounting for them widens

[1] Because each person is observed for only a year, reassessments that occurred before a person entered the survey are not observed.

measured wage differentials. This finding that conditions are part of compensation is important for this analysis, which explores whether a shock changed the way workers value the nonpecuniary aspects of their job.

Two further papers show that the weight placed on job characteristics is not the same for everyone. Mas and Pallais (2017) find that women, particularly those with young children, are willing to give up more pay than men to work from home and to avoid schedules set at the employer's discretion, and Wiswall and Zafar (2018) find that women place a higher value than men on work flexibility and job stability, suggesting it is important to differentiate the way policies impact employment outcomes by gender.

B. The Labor Market in the Pandemic Recession

Payroll employment fell by 22.4 million in March and April of 2020, a decline of 15 percent from the February peak, with losses of 1.7 million in March and 20.7 million in April, and the unemployment rate reached 14.8 percent in April, the highest since the series began in 1948 (Ansell and Mullins 2021; Bureau of Labor Statistics, Employment Situation). The speed of the decline was without precedent, as was also the speed of the partial recovery that followed. The April decline alone was the largest in the history of the CES payroll employment series, which begins in 1939, and the two-month fall compares with job losses averaging 3 percent across the four previous employment downturns since 1981, the largest of which was the 6 percent lost in the Great Recession (Ansell and Mullins 2021).

What distinguishes the episode for our purposes is not the magnitude, but which workers were disproportionately affected. In previous recessions the losses fell most heavily on goods production and construction. This time they fell on contact-intensive services. For example, accommodation and food services, which are part of the leisure and hospitality industry, lost 6.9

million jobs in the two months following its February peak. This comprised most of the 8.2 million or 49 percent decline in jobs in leisure and hospitality in March and April, which was the largest of any major industry group. The contrast inside food service illustrates the mechanism as employment at full-service restaurants, where patrons order and are served while seated, fell 65 percent, while employment at limited-service restaurants, which rely on takeout, drive-through and delivery, fell a smaller 23 percent.

In general, Ansell and Mullins (2021) report that job losses over 2020 were greatest in the industries in which employees, customers, or both come into close contact. They are also industries that employ women heavily, so the demographic incidence of the recession followed from its industrial incidence. Women accounted for 12.2 million of the 22.4 million jobs lost in March and April, 55 percent of the total, against a 50 percent share of February employment. Women were 53 percent of employment in leisure and hospitality and 77 percent in private education and health services, and within those industries the losses fell disproportionately on them, as women were 84 percent of the education and health decline and 54 percent of the leisure and hospitality decline (Ansell and Mullins 2021).

How much of that loss was caused by the policy response rather than by the virus itself is contested. Gupta et al. (2023) find that the employment rate fell by about 1.7 percentage points for every additional ten days a state spent under a stay-at-home mandate between mid-March and mid-April 2020, with the adjustment coming almost entirely through job loss rather than reduced hours or earnings, and Beland, Brodeur, and Wright (2020) find unemployment effects significantly larger in states that implemented such orders. Others find little impact of these orders. Lin and Meissner (2020) report that job losses were no higher in states that implemented stay-at-home orders than in states that did not, and Chetty et al. (2020) find that spending fell

sharply in mid-March 2020, led by high-income households in areas with high infection rates, and that state-ordered reopenings had little effect on local employment. As the impact is uncertain, we include policy measures in our model as controls so that our estimates of the importance of job quality are not confounded by differences in pandemic policies.

The recovery in the aggregate labor market was rapid, but it was not uniform and it took time to get back up to trend. Higgins, Pitts, and Wiczer (2024) show that, as of September 2024, although payroll employment was above its pre-pandemic level, a gap of about 3.4 million jobs remained relative to its pre-pandemic trend. Growth had come largely from health care and social assistance, government, and leisure and hospitality. The first two were approaching their pre-pandemic trends, but leisure and hospitality, on the evidence of vacancies and vacancy yields, was not.

C. Unemployment Insurance and Extended Benefits

Part of the policy response ran through unemployment insurance. The CARES Act of March 2020 created the Pandemic Unemployment Assistance (PUA) program, which extended Unemployment Insurance (UI) eligibility to workers not normally covered, the Pandemic Emergency Unemployment Compensation (PEUC) program, which extended the duration of benefits, and the Federal Pandemic Unemployment Compensation (FPUC) program, which added a federal supplement of $600 a week through July 2020. The supplement was renewed at $300 a week from late December 2020 through early September 2021 (Ganong, Greig, Noel, Sullivan, and Vavra 2022). Alongside these, the long-standing Extended Benefits program was triggered in most states as their unemployment rates crossed the statutory thresholds, lengthening the period over which a worker could claim. That trigger is a state-level policy that turned on and

off at different times in different places, which makes it useful in identifying workers' pandemic policy exposure.

These supplements were large relative to prior earnings. Ganong, Noel, and Vavra (2020) find that 76 percent of workers eligible for regular unemployment compensation had statutory replacement rates above 100 percent between April and July of 2020, with a median replacement rate of 145 percent. An obvious concern was that benefits of this size would keep workers from returning to work. However, the evidence on that concern is largely negative. Ganong, Greig, Noel, Sullivan, and Vavra (2022), using administrative bank account data, find large spending effects and small job-finding effects. Marinescu, Skandalis, and Zhao (2021) find that the supplement reduced search effort without reducing vacancy creation and argue the employment effect was muted because jobs were unusually scarce at the time, so fewer applications had little effect on hiring. Dube (2021) finds no association between states' relative replacement rates and job growth around the expiration of the supplement.

That literature asks whether enhanced UI benefits kept workers out of employment. It does not ask whether benefits changed what workers were willing to do. These subsidies could potentially allow a worker to be more selective about the job amenities as well as the wage. We therefore include the Extended Benefits trigger alongside the closure and mandate measures as a mitigating qualifier on pandemic exposure. It is a state-level shift in the value of not working, and it varies in a way that can be separated from any shutdown orders.

D. Labor Shortages and Wage Growth

From the employer's side, the recovery looked like a shortage. The number of job openings reached a series high of 12.3 million in March 2022, the highest since the Job Openings and Labor Turnover Survey (JOLTS) began in December 2000 and quits matched the series high of

4.5 million in the same month. Both were concentrated in the low-wage, contact-intensive industries that had lost the most employment. In September 2021 the overall quits rate reached 3.0 percent, the highest since the series began, and the industry pattern is stark, at 6.4 percent in leisure and hospitality and 6.6 percent in accommodation and food services, against 3.4 percent across total private employment (Bureau of Labor Statistics, Job Openings and Labor Turnover Survey).

Wages responded, and they responded most at the bottom of the wage distribution (for example, see Hotchkiss 2021). Autor, Dube, and McGrew (2023) document a compression of the wage distribution driven by young non-college workers moving from lower-paying to higher-paying employers, which they attribute to low-wage employers facing more competition for workers. As workers became more willing to move for higher pay, those employers had less power to hold wages down. The gains came mostly through workers changing jobs rather than through raises in the same job. The Atlanta Fed's Wage Growth Tracker shows the same pattern, with the gap between job switchers and job stayers reaching a series high of 2.7 percentage points in November 2022.

Part of that adjustment was explicitly about amenities rather than pay. Barrero, Bloom, Davis, Meyer, and Mihaylov (2022) show that the shift to remote work raised the amenity value of employment, and that as compensation adjusts to share those gains with employers it moderates wage-growth pressure. By their estimate, the added amenity value of remote work took the place of about two percentage points of cumulative wage growth over two years; without it, wages would have risen that much more. That is evidence that a change in job characteristics shows up in the wage data, and it applies to a characteristic available mainly to workers who were already well paid.

This reallocation of workers during the recovery could be explained by workers changing preferences or taking advantage of opportunity. Fee (2022) finds that mobility out of low-quality jobs rose substantially from 2010 to 2019, contracted in 2020, and returned to pre-pandemic rates in 2021 as conditions improved, concluding that tight labor markets are what let workers move into better jobs. On that reading nothing about preferences changed; opportunity did. Similarly, Forsythe, Kahn, Lange, and Wiczer (2022) find that early dynamics were driven by temporary layoffs and recalls, the remaining shortfall by 2022 reflected retirements, and the reallocation across jobs was no greater than in years prior to the pandemic. Hodge (2022), working with related data, documents which roles workers were leaving during the "Great Resignation" and finds occupations entering the top of the list that had not appeared there before. Retail occupations sit near the top of his list in all three periods. Registered nurse became a top occupation that workers left for the first time in 2020-21, and elementary and middle school teacher rose from eighth place in 2018-19 to fifth in 2019-20 and sixth in 2020-21. Nurses and teachers were among workers experiencing the worst of degrading working conditions during the pandemic.

Pay and job characteristics are correlated in the cross section as the jobs that had to raise wages most to retain or attract workers during the recovery are also the jobs in which a worker has least control over their day and most contact with the public. If the reallocation were only about opportunity, the destination of a move should be predicted by pay and by the state of demand, and the characteristics of the origin job should add nothing once wages and demand are held constant. Distinguishing the two therefore requires measuring the characteristics directly rather than inferring them from pay.

E. Evidence That Workers Left Risky Work

The closest analysis to ours is Braakmann, Eberth, and Wildman (2022), who find that wages did not adjust but that workers began leaving high-risk occupations during 2020, with larger effects among those not covered by lockdown or work-from-home provisions. Their finding that workers moved while wages (on the job) did not is the pattern we also expect, and why quantifying flows of workers, rather than merely wage changes, is important.

However, workers' options in the United Kingdom were quite different as the U.K. operated a furlough scheme through most of 2020 and 2021 while the United States did not. A furlough holds the employment relationship open, so the margin between leaving a job and leaving an occupation sits in a different place there than in the United States. A British worker who wanted out of a risky occupation could wait on furlough, while an American worker in the same position was more likely to have separated already and faced the question of what to return to.

F. Employment, Children, and the Household

In addition to those who took advantage of the tight recovery labor market to improve their working conditions and/or wages, there were many workers who left the labor market entirely. Who left employment and did not come back was not evenly distributed across the workforce. Pitts (2021) finds that the decline in employment fell disproportionately on women, unlike past recessions, and that the presence of young children was a driving factor in that differential. Compared with the Great Recession and with the 2019 expansion, women with young children, particularly single women, showed a markedly lower attachment to employment during the pandemic than men or women without children. Women with children under age six made up about 10 percent of the pre-pandemic workforce and accounted for almost a quarter of the unanticipated, COVID-related decline in employment. Alon, Doepke, Olmstead-Rumsey, and

Tertilt (2020) also make this point directly, noting that, unlike ordinary recessions which fall harder on men, the employment drop associated with social distancing fell on sectors with high female employment shares, and closures of schools and daycare centers raised childcare needs in a way that hit working mothers hardest.

A parallel literature asks what school closures in particular did to parental employment, and its findings bear directly on the child age bands we estimate. Landivar et al. (2023) show that employment losses were concentrated among mothers of young school-age children rather than mothers of preschoolers. Between 2019 and 2020 employment fell 7 percent among mothers whose youngest child was five to twelve, against 4.8 percent where the youngest was four or younger, and by 2021 the gap had widened, with losses of 5.4 percent for the school-age group against 2.9 percent for mothers of the youngest children. They attribute the difference to school closures having been more extensive and more prolonged than closures of formal and informal childcare, so that mothers of preschoolers lost less of the care they had been relying on. They also find that remote instruction widened the employment gap between mothers and fathers by as much as 5 percentage points, and that the disparities endured through spring 2021, after many districts had returned to in-person instruction. Related work using the same school operating status data reaches similar conclusions about the gender gap (Collins et al. 2021; Landivar et al. 2022). These results suggest that the capacity to act on a preference about job quality is not the same for everyone. Leaving a job for a better one, or declining to return to a worse one, requires the ability to absorb any gap in pay, and that ability is a property of the household rather than of the worker.

III. Data

We draw on multiple sources for the analysis, including the Current Population Survey (CPS), data on occupational characteristics from the Occupational Requirements Survey (ORS) and O*NET, and several data sets on public policy and business or school closures during the pandemic.

A. Current Population Survey

Our primary data source is the CPS, whose basic monthly microdata we retrieve from the Census Bureau (Census Bureau, Current Population Survey). The CPS is a monthly survey of about 60,000 households conducted by the Census Bureau on behalf of the Bureau of Labor Statistics (BLS) that provides a snapshot of the condition of the labor market each month. The CPS relies on continuous monthly interviewing under a 4-8-4 rotating panel design. A household is interviewed for four months in a row, then is out of the sample for eight months, and then is interviewed again for another four consecutive months. This allows us to see the characteristics of a person's job at two dates a year apart, when the person is employed at both. We investigate, for workers who changed occupation, how occupational characteristics changed from one year to the next in the years surrounding the COVID-19 pandemic.

The CPS's rotating panel design means that a household in its fifth through eighth interview was also in the sample twelve months earlier. We keep those records and match each worker to their own record a year earlier, requiring that gender and race agree and that age advance by no more than one year. Records missing an age group, an education group, gender or race are dropped, as is Puerto Rico. The final file includes the last four months of interviews for the years 2018 through 2025, with 1,601,379 person-month observations. Each person contributes up to

four records, one for each of the fifth through eighth interviews, and the twelve-month windows of those records overlap.[2]

All 1,601,379 matched person-months are of workers employed in period t-12. Of these, 90.4 percent are also employed in period t, and of those employed at both dates, 29.5 percent record a validated occupation change (a process described below). Across the three periods, January 2018 through February 2020, March 2020 through December 2022, and January 2023 through March 2025,[3] the share employed at both dates is 91.3, 89.1 and 91.2 percent, and the share recording a validated occupation change is 29.0, 29.8 and 29.7 percent. The pandemic lowered response to the CPS, from an average of 83 percent in the year before March 2020 to 65 percent in June 2020, recovering to 77 percent by July 2021 (Bureau of Labor Statistics, Employment Situation COVID-19 FAQ). Lower response reduces the number of records that can be matched across the year and may change who is matched during the pandemic period; the controls hold observed characteristics fixed, but a shift in the unobserved characteristics of those who responded would not be captured.[4]

One of the key variables in this analysis is occupation switching. The reliability of occupation changes is an ongoing issue with the CPS. Since 1994 the CPS has used dependent coding, so in interviews other than the first and fifth, the respondent is reminded of the job reported the month before and asked whether it has changed. A worker who reports no change keeps the previous

[2] The CPS samples addresses rather than people, so a household that moves is not followed and a person's records always fall in the same state. For the policy measures, clustering on state therefore allows for any correlation among a person's records as well as for the common policy shocks that records from the same state and period share.

[3] March 2025 is the last month for which the QCEW industry employment data used for the demand shock were available when the sample was built.

[4] The estimates are unweighted. The survey weights are built to make each monthly sample represent the population, and the two records compared for each person carry different weights that change between the two dates, so no single weight applies to a matched pair. Our aim is to estimate how outcomes respond to the characteristics of a worker's job, not to describe population totals, and the models condition on age, sex, race and ethnicity and state, the characteristics the weights adjust for; in that case weighting is not needed for consistent estimates and would reduce precision (Solon, Haider, and Wooldridge 2015). The estimates therefore describe the matched sample.

month's occupation code, and only a worker who reports a change is asked to describe the new occupation, so month-to-month code changes reflect reported job changes rather than differences in how the same job was coded. Dependent coding does not link the two records we compare, however. Every comparison in our sample spans the eight-month break in the rotation, since each record in its fifth through eighth interview is compared with the record twelve months earlier, so the two occupation reports are coded independently, and an apparent change may reflect different coding of the same job (Moscarini and Thomsson 2007; Kambourov and Manovskii 2013). We therefore rely on additional evidence of a change, including a change in worker class (private, government, incorporated or unincorporated self-employed, and work without pay), a change in industry, or looking for work in the last 12 months, adapting the filters Moscarini and Thomsson (2007) use to validate month-to-month occupation changes. If a switch did not meet these criteria, then it is treated as no change in occupation. Occupation is measured with the harmonized 2010 Census occupation codes provided by IPUMS-CPS (Flood et al. 2025), which are merged onto the Census records using individual identifiers, so the move of the CPS to the 2018 Census occupation classification in January 2020 does not by itself create a change in code. We test the implications of this validation decision in Section V.D.

B. Occupation Characteristics

The ORS is a survey of employers undertaken by the BLS, in partnership with the Social Security Administration, that provides the employer's perspective on the "…physical demands; environmental conditions; education, training, and experience; as well as cognitive and mental requirements for jobs in the U.S. economy" (BLS 2026). The ORS is organized into waves. The first wave was collected between 2015 and 2018, and the second as five sample groups fielded in successive years. This analysis uses the reference year 2023 estimates, which are the final

estimates for the second wave and pool all five sample groups, collected between September 2018 and 2023. Because occupations are not resampled every year, only the completed wave provides estimates for the full set of occupations, which is why the 2023 reference year is used here.[5]

While the ORS provides the employer's perspective, the worker's perspective on the characteristics and requirements associated with occupations is captured by O*NET, which is a survey of workers undertaken by the Employment and Training Administration of the U.S. Department of Labor. We use the Work Context items of the May 2025 release, version 29.3 (National Center for O*NET Development 2025).

Characteristics from both the ORS and O*NET attached to occupations are used to construct two occupational amenities, autonomy and safety. Details of how these measures are constructed are contained in a separate online appendix. Briefly, six autonomy-related and seven safety/exposure-related variables were selected from the two surveys, reflecting aspects such as control over work, schedule regularity, public interaction, and exposure to crowds or disease. Principal Component Analysis (PCA) was applied to these standardized variables across 474 occupations (weighted by employment), reducing each set to its first principal component, which served as the summary measure for autonomy and safety. The autonomy score combined items like sit-stand choice and self-paced work, while the safety score emphasized absence of public contact and exposure, with telework carrying minimal weight.

[5] The first wave, collected between September 2015 and July 2018, published final estimates for far fewer occupations than the second (347 against 477 SOC codes) and classified them on the 2010 SOC, so it cannot supply scores for the full set of occupations in our sample. The O*NET measures are taken from the May 2025 release of the O*NET database (version 29.3), whose data collection falls in a period comparable to that of the second ORS wave.

Figure 1 plots the autonomy score and the safety score for each broad occupation. The occupations of higher "quality" based on the characteristics included in their construction are what we would expect. Relative to Management, Business, Science and Arts occupations, all others have a lower "quality" in the autonomy dimension, as they all have lower levels of the ability to pause, self-pace, flexibility in sitting/standing, autonomy regarding decision making, self-determination on job tasks, and, for most, the ability to count on a regular schedule. The occupations with the lowest autonomy scores are Food Preparation and Cleaning and Transportation and Material Moving.

Regarding the quality of occupations as they relate to exposure characteristics, occupations that could be characterized as traditionally blue-collar occupations have the highest safety score. Even though these occupations are among the lowest for teleworking options, workers in these occupations do not have to contend with crowds, with interacting with the public or with angry people, or with exposure to infectious conditions. Those in Protective Service, Education and Health Care, or Sales have the lowest safety scores.

We assign both scores at the four-digit occupation level to the occupation the worker held twelve months earlier. The scores are measured once and carry no year subscript, so an occupation's autonomy and exposure are the same in 2018 as in 2023. We interpret a coefficient that moves across the periods as a change in how workers weighed a characteristic, although it could also reflect changes in the demand for those occupations or in who holds them. The ORS estimates pool collection from September 2018 through 2023 and O*NET ratings are collected over several years, so the scores describe each occupation over roughly that span rather than at a point in time; because the items measure persistent features of the work, we expect them to embed little of the pandemic’s temporary changes. Holding the scores fixed also means that

identification comes from differences across occupations within a state and month, so that two workers under the same mandates and the same local conditions differ in what their jobs entail.

C. State and Local Pandemic Policy

We use multiple sources to identify the state and local COVID policies that may have affected both employment and occupational switching. Each source records the dates on which an order was in force, and, as a single indicator dated at the interview would be misaligned with the interval over which the decision was made, we build a measure of the share of days in a window that an individual in that state would have spent under the order. The measure runs from zero to one, and an estimated coefficient reflects the difference between a window spent entirely under the order and one spent entirely outside it. Several of the sources end before our sample does. Wherever that is so, a day after the source's last date is coded as not under the order, whether or not a policy was in force, and the end dates and what they mean for each measure are given at the end of this section.

The window runs from twelve months before the CPS interview through the month before it. We exclude the interview month itself, because the occupation is already recorded by then. Because the occupation change happened at an unknown point inside the window, we split the window into two halves, with a distant half covering months twelve through seven before the interview, and a recent half covering the six months immediately before it. Workers also change occupation at different times, and some change more than once. Only the occupation held at the second date is observed, so the destination we measure is that of the most recent move. A policy in force in the recent half is therefore closer in time to that move than one in force earlier, although the move itself can fall in either half.

The data are merged with the CPS at the lowest possible level of geography. For confidentiality, the CPS identifies only larger counties. Thus, we matched the pandemic policies and closure data in a hierarchical manner for county level data. If the county was identified in the CPS, the data were matched at the county level. If the county was not identified in the CPS but the CBSA was, that match was used, using the average of the policy information for the counties not identified but known to be in the CBSA. If a county was not identified in either, we used the average policy value for the remaining counties.

Data on mask and vaccine mandates come from two Centers for Disease Control and Prevention (CDC) datasets, the county-by-day public mask mandates of April 10, 2020, through August 15, 2021 (CDC 2021) and the state-level vaccine mandates issued from July 26, 2021, as last updated on September 25, 2022 (CDC 2022), from which we compute the number of days each mandate was in force in each month. There are intra-state variations in mask mandates for only two states, New Mexico, which has a match between some of the counties with differential mask mandates, and Mississippi, where mandates varied by county but none of these were identified in the CPS, so the value is the share of the state's population under a mandate. The vaccine data are state-level requirements covering education, government, healthcare and the general public. We use the share of days under any mandate in each of the two six-month periods, taking the largest of the four groups' days in each month rather than the sum, so that a state running two overlapping mandates is not counted twice. In industry-specific estimations we substitute the group-specific measure, the education mandate for the education group and the healthcare mandate for the health care group, since for those workers the mandate is the treatment rather than a proxy for it.

Data on business closures and stay-at-home orders come from the COVID-19 US State Policy Database (Raifman et al. 2022), which records state-wide directives, with the dates each was in force. We treat closure as a composite with the order counted as in force on any day on which any of five orders applied. The orders cover five different types of businesses, including nonessential businesses, bars, restaurants, gyms, and hair and nail salons. Bar closure accounts for 98 percent of the composite, and in 5.7 percent of the sample bars or restaurants were closed while nonessential businesses were not, which is why the composite rather than the nonessential-business order alone is used. Stay-at-home orders are carried separately. For both the closure composite and the stay-at-home order, we use the share of days the order was in force in each of the two six-month periods.

The COVID-19 US State Policy Database (Raifman et al. 2022) also provides the availability of Extended Benefits, the program that lengthens the duration of unemployment insurance when a state's unemployment rate is high enough. We include it because a state that activated it changed the value of not working. Because the program switches on when a state's unemployment rate rises, its coefficient also partly reflects local labor market conditions. The federal pandemic programs (FPUC, PUA and PEUC) were national and are absorbed by the year-month effects. We use the share of days the program was active in the two six-month periods.

Data on school closure come from Parolin and Lee (2021), who measure closure from anonymized mobile-phone foot traffic at more than 100,000 public schools, comparing visits in a month with the same month a year earlier. We use the closure rate, the share of local public schools that were closed, for the school year in which the record falls, taken as the mean of the first and fourth quarter rates, as differences in school start and end dates for summer break vary

greatly across the country. A second indicator marks whether the record falls in one of the two school years the closure data cover, 2020-21 or 2021-22, so that a closure rate of zero outside those years is not read as schools being open. Unlike the other policy measures, the closure rate is assigned by the school year in which the record falls rather than by the share of days in each half of the window. As there is limited closure data in the first quarter of 2020, these closures are not explicitly captured in the data. However, closures were nearly universal and thus are absorbed by the year-month effects.

Several of the policy sources end before our sample does, and after each end date the measure is recorded as zero whether or not a policy was in force. The CDC mask mandate data end on August 15, 2021, so mandates in force after that date are recorded as zero, and the mask measure understates exposure in the second half of the pandemic period. We use the version of the COVID-19 US State Policy Database released on March 30, 2022. The stay-at-home orders and business closures it records had ended before January 2022, the earliest month in the window of any post-period record, so those measures are zero after 2022. Extended Benefits are recorded through March 2022, and any activation after that date is recorded as zero; few states had Extended Benefits in effect after that date, among them New Jersey, whose Extended Benefits period ended on April 9, 2022 (New Jersey Department of Labor and Workforce Development 2022), so these zeros are largely accurate. A vaccine mandate issued after September 25, 2022, is not recorded, but mandates in force at that date remain so until the expiration date coded in the law, so vaccine mandates extend into the post period. Because each measure covers the twelve months before the interview, measures that end in 2022 can still take nonzero values for records interviewed in early 2023.

D. Occupation and Industry Pay

The CPS reports earnings only for the outgoing rotation groups; thus, in a given month, there is no individual wage for approximately 75% of the observations. Therefore, we use the mean pay of the occupation and industry rather than the worker's own wage. We capture pay with four variables, a level and a deviation for both the occupation and the industry. The first is measured at its 2016-17 value, in 2017 dollars, for the four-digit occupation, with a fallback to the two-digit and then the one-digit group mean where a cell is thin, and for the broad industry. Fixing the level before the pandemic is what allows it to control for the kind of job a worker started in without absorbing the pandemic pay movements that are themselves of interest.

The second is a deviation, the proportional gap between the occupation's pay over a window and that same 2016-17 level. Both are in 2017 dollars, so the deviation measures real pay growth. This allows us to capture occupation and industry wage growth. We compute it over the same two six-month halves used for the policy measures, from raw monthly pay rather than a smoothed series, so that the halves carry separate information. For occupation the two halves correlate at 0.221, but at the industry level they correlate at 0.768, which is too close to separate. Thus, the industry deviation enters as a single twelve-month window.

E. Outside Options

Whether a worker leaves an occupation depends not only on the job they hold but on what else is open to them. Furthermore, what is open to them is likely not the whole economy but the set of occupations that people in their position actually move into. We build that set from the observed transitions out of each origin occupation, weighting destinations by the number of workers who entered them and leaving the respondent's own transition out of the calculation. Leaving the respondent out means the measure is defined for workers who stayed as well as those who

moved and is not a mechanical consequence of the respondent's own decision. Origin occupations at the four-digit level with fewer than twenty leavers use the two-digit occupation measure.[6]

We have five different measures of outside options. Two are the average gain in the autonomy score and in the safety score, destination minus origin, among workers who left the origin occupation, on the same scales as the origin scores. These are computed on annual rolling windows and are well measured, with reliabilities of 0.967 and 0.975. We rely mostly on the rolling measures in the analyses because they describe the options workers actually faced in each year, relative to the occupation they held. Leaving the respondent out removes the worker's own contribution to the measure but not a shock common to the origin and destination occupations, which is why equation 3, where such a shock would bear most directly on the outcome, holds the outside options at their pre-2020 values. There are three outside-option measures of pay. The first is the pay gap, which is the 2016-17 pay of the destination occupations minus that of the origin occupation, in dollars an hour. The second is the pay deviation gain, which measures how far the destination occupations' pay over the preceding twelve months stood above their 2016-17 level, relative to the same measure for the origin occupation. The third is the switcher wage premium, which is the mean individual wage growth of workers who changed occupation minus that of workers who stayed, among workers from the same two-digit origin group in the same period, measured from the wages reported in the outgoing rotation groups. The pay measures are computed over the three periods rather than annually, because wages are only collected from outgoing rotation groups and at the four-digit occupation by year the sample size is too small.

[6] About 3 percent of records use the two-digit measure; they are not separated out in the estimates. Reliability is one minus the mean squared standard error of the occupation means divided by their variance, which is the share of the variation across occupations that is signal rather than sampling noise. At these values, sampling noise in the measures is too small to bias their estimated effects materially toward zero.

F. Industry Labor Demand

The closure orders fell hardest on industries that are themselves low-autonomy, so without a control for demand the quality-by-period interaction could be picking up collapsing demand rather than any reassessment of amenities. We therefore measure the demand conditions the worker's industry faced. For each industry we compute the year-over-year growth in national employment, excluding the worker's own state from the calculation, and assign it by the industry the worker held twelve months earlier. Employment counts come from the Quarterly Census of Employment and Wages (QCEW) by state and supersector for 2014 through 2025 (Bureau of Labor Statistics, Quarterly Census of Employment and Wages). Excluding the own state means the measure is not driven by the same local conditions that determine the worker's own outcome. It is the shift alone rather than a shift-share, and it enters both equations, since industry demand plausibly bears on whether a worker stays employed as well as on whether they change occupation.

G. Spouse and Household

For individuals for whom a spouse or partner is present we use their CPS record to create the same occupation measures we use for the worker, including the autonomy score and safety score, the pay of the occupation they hold at its 2016-17 level, and the pay deviation over each half of the window, in addition to the employment measure. Where no spouse is present these measures are set to zero and four indicators capture whether the worker is married or partnered, whether the spouse is employed, whether the spouse has no occupation, and whether the spouse’s industry cannot be determined. The spouse's autonomy score and safety score enter both equations as household controls. The spouse's job may shape how the pandemic affected the household, including whether the spouse's own employment was at risk, how much exposure the

spouse brought home, and so whether the worker was able or willing to stay in work or to change jobs.

H. Demographics and Remaining Controls

We control for the worker's gender, age and its square, education, race and ethnicity, and the presence of children in three age bands, under six, six to twelve, and thirteen to eighteen. Gender is interacted with the child indicators, as the impact of the pandemic has been found to differ by gender and the age of the child (Pitts 2021). The model also includes industry and metropolitan status indicators as well as state and year-month fixed effects.

I. Sample Characteristics

Table 1 reports the means of the variables used in the models, for the full sample and for each period. Of the workers employed at the first date, 90.4 percent are also employed a year later, a share that is lower during the pandemic, at 89.1 percent, than before or after it. Among those employed at both dates, 29.5 percent change occupation, and that share barely moves across the periods. The average autonomy score of the occupation held rises from 0.25 before the pandemic to 0.35 after it, while the average safety score changes little. Occupation pay rose in real terms relative to its 2016-17 level over the period, with the recent pay deviation growing from about 1 percent before the pandemic to about 5 percent after it. The policy measures are zero outside the pandemic period, except for vaccine mandates and Extended Benefits, which carry into 2023. During the pandemic, the average worker lived in a state with a mask mandate in force on about a quarter of the days in each six-month half, and over the 2020-21 and 2021-22 school years, the two the closure data cover, the average share of local public schools closed is 24 percent. The composition of the sample is otherwise stable across the periods. Women are 48 percent of the records, and the average age is 46, about two-thirds of workers are married or partnered and

about half have an employed spouse, and the share with a child in each age band changes by less than a percentage point. The share with a bachelor's degree or more rises from 40 percent before the pandemic to 43 percent after it, and the share of Hispanic workers from 13 to 14 percent.

IV. Model

To examine how job quality and the pandemic affected employment outcomes, we estimate two probit models. The first determines whether the worker was employed at both ends of the twelve-month interval and is estimated on all matched records employed at t-12. The second models occupational change and is estimated on those employed at both dates. The two equations are estimated separately rather than jointly, for the reasons set out below. The employment equation, for individual i in state s and time t is:

$$E^*_{ist} = \alpha_1 C_{i,t-12} + \alpha_2 X_{i,t-12} + P'_{st}\alpha_3 + \alpha_4 B_{j,t-12} + W'_{it}\alpha_5 + \theta_s + \tau_t + \varepsilon_{ist} \text{ , (1)}$$

with the worker employed at both dates when $E^*_{ist} > 0$, which is conditional on being employed in period *t-12*. The switching equation is observed only for those employed at both dates and takes the form of:

$$S^*_{ist} = \beta_1 C_{i,t-12} + \beta_2 X_{i,t-12} + P'_{st}\beta_3 + O'_{i,t-12}\beta_4 + \beta_5 B_{j,t-12} + W'_{it}\beta_6 + \theta_s + \tau_t + u_{ist} \text{ , (2)}$$

where $C_{i,t-12}$ and $X_{i,t-12}$ are the autonomy score and the safety score of the occupation held twelve months earlier, P_{st} are the policy exposure shares, $O_{i,t-12}$ are the outside-option terms, which enter the occupation equation only, $B_{j,t-12}$ captures the demand shift in the worker's industry j, W_{it} are the demographic and household controls, which include spousal employment information, and θ_s and τ_t are state and year-month fixed effects. The three periods are pre, January 2018 through February 2020; during, March 2020 through December 2022; and post, January 2023 through

March 2025. The error terms ε_{ist} and u_{ist} are not restricted to be jointly normal, and no correlation between them is imposed or estimated. Each equation is fitted as a single-equation probit by maximum likelihood. Because the policy measures are identical for every worker in a state and month, while the job quality scores take one value for each of the 474 four-digit occupations, no single choice of cluster serves both sets of variables. We therefore judge each variable with standard errors clustered at the level at which it varies, on state for the policy measures and on the origin occupation for the quality scores, occupation pay and the outside options. The figures show confidence intervals on the same basis. The online appendix reports the coefficients of every model of equations 1 and 2 with standard errors clustered on state and on occupation, and repeats every figure with standard errors clustered on state.[7]

The coefficients of main interest are those on the interactions of the autonomy score and the safety score with the three periods, added in the period model described below, which govern how the effects of the two scores differ across the periods in each equation. If the pandemic changed what workers want from their jobs, the weight the switching decision places on the quality of the origin occupation is expected to differ across periods, with workers starting in low-autonomy or high-exposure occupations being more likely to leave them during the pandemic than before it.

The two equations are estimated separately. A bivariate probit with sample selection would correct the switching estimates for non-random selection into employment at both dates, and that

[7] Two-way clustering on state and occupation is not available for a probit in Stata. In a linear probability model of the same specification, where it is available, two-way standard errors fall within one percent of the occupation-clustered ones for every job quality and pay measure, and recover the state-clustered values for the state-level policy measures, so two-way reproduces whichever of the two single clusterings is larger, term by term. Clustering on occupation rather than state raises the standard errors on the quality scores by a factor of six to ten on the occupation margin and three to four on employment at both dates, and lowers those on the policy measures by between a quarter and a half. Online Appendix Table OA10 reports the period model with the outside options fixed at their pre-2020 values, and Tables OA1 to OA9 report every model with both sets of standard errors.

correction is credible when some variable shifts the employment margin without entering the switching decision. No variable available to us does both. The mean pay of the occupation held by the worker's spouse or partner predicts women's employment at both dates strongly in every period but, entered directly in the switching equation, it is also significant for women ($p = 0.029$). A leave-one-out employment demand shock in the spouse's industry predicts employment at both dates pooled across the periods but not within every period (Appendix Table C1).

Fitting the joint model anyway would identify the correlation between the two equations from the assumption of bivariate normality rather than from the data. In Heckman's (1979) selection model the correction term is a nonlinear function of the selection index, so the model is formally identified even without an exclusion. Puhani (2000) and Bushway, Johnson, and Slocum (2007) explain why that is not a sound basis, since the correction term is close to linear over most of its range. Without a credible exclusion the correction offers no improvement over uncorrected estimates and can increase bias. Appendix C reports the joint model, together with the evidence that its estimated correlation does not respond to which variable is excluded.

Both spouse variables are retained in both equations as household controls. The spouse's industry demand shock is constructed in the same way as the worker's own, as the year-over-year growth in national employment in that industry with the worker's state excluded.

The switching results describe the probability that a worker who was employed in both periods switched occupations between *t-12* and *t*. Selection is limited in scale, as 90.4 percent of workers employed at the first date are also employed at the second, so the occupation-change estimates describe most of the original sample, and the concern is whether the 9.6 percent who are not differ in ways the model does not capture. The share employed at both dates falls from 91.3

percent before the pandemic to 89.1 percent during it, so selection bears somewhat more on the pandemic-period estimates. The joint model of Appendix C estimates a correlation of -0.41 between the unobserved determinants of the two equations, identified only from functional form, and its occupation-change estimates are nearly identical to those reported here; that agreement is consistent with small selection bias but, given how the correlation is identified, is not evidence against it. Thus, we are unable to extrapolate to workers who left employment.

We estimate equations 1 and 2 in four forms. The first is equations 1 and 2 as written, estimated on the full sample, which gives the average response across all workers. The second interacts the two quality scores with the three periods, with each other, with occupation pay and with the policy measures, and interacts the outside options with the periods; these ask whether the weight a worker placed on the quality of their job depended on what it paid, on what policy was in force, and on when the decision was made, and whether what was available elsewhere came to matter more. The third interacts gender with every regressor, including the state and year-month effects, in a single model, so that no coefficient is constrained to be common across women and men and every difference between them can be tested. Within the model the presence and ages of children are interacted with the policy measures, the quality scores and the outside options, so that the household enters wherever it could plausibly operate. The fourth re-estimates the first specification separately within groups defined by age, education, broad industry and occupation pay, and asks where in the workforce the response, if any, is concentrated.

Note, the switching equation records whether a worker changed occupation, not whether the new occupation was better than the old one. We do not classify moves as better or worse, for three reasons. First, any classification needs a threshold for how large a change in score counts as

a move up or down, with smaller changes treated as lateral. The scores have no natural unit, so the threshold is arbitrary, and the share of moves counted as up or down depends on where it is set. Second, where a worker starts limits where they can go. A worker in one of the highest-scoring occupations can barely move up and one in the lowest-scoring can barely move down, so the origin score predicts the direction of a move mechanically rather than behaviorally. Third, autonomy and safety are separate dimensions, and a move can raise one while lowering the other, so a classification on either dimension alone cannot say whether the new job was better overall.

We therefore measure direction continuously. Among workers with a validated occupation change, we estimate the change in each score between the occupation held at t-12 and the one held at t, by least squares, on the same regressors as the period form of equation 2:

$$\Delta Q_{ist} = \gamma_1 C_{i,t-12} + \gamma_2 X_{i,t-12} + P'_{st}\gamma_3 + O'_{i,t-12}\gamma_4 + \gamma_5 B_{j,t-12} + W'_{it}\gamma_6 + \theta_s + \tau_t + e_{ist} \quad (3)$$

where ΔQist is the change in the autonomy score or the safety score, estimated separately for each, and the remaining terms are as in equation 2, except that τt here denotes calendar-month effects together with period effects, because year-month effects would absorb the differences across periods that equation 3 is estimated to measure. No threshold is needed, and a move that raises one score while lowering the other is recorded as it is on each. The origin scores are in the model and interacted with period, which absorbs the mechanical relationship between where a worker starts and how far they can move; for the same reason their own coefficients are not interpreted. What is interpreted is how the adjusted mean change differs across the three periods. The question is whether, holding composition fixed, workers who changed occupation moved further toward autonomy and safety during and after the pandemic than before it. Like the

switching equation, equation 3 describes workers employed at both dates and cannot be extrapolated to those who left employment. In equation 3 the outside-option measures are held at their pre-2020 values for each origin occupation, because the rolling measures used in equation 2 are close to an average of the outcome of equation 3 itself and could absorb the shift it is meant to detect (Section V.C).

V. Results

This section reports employment at both dates first, followed by occupation change. For each outcome we report the main model, the model that lets the response differ across the three periods, the gender model and the other heterogeneity estimates; for occupation change we also report the change in job quality among workers who changed occupation. The average marginal effects are in percentage points, against base rates of 90.4 percent for employment at both dates and 29.5 percent for occupation change. Continuous measures are reported for an interquartile change, the difference between the occupation at the 25th and at the 75th percentile of the measure, which is 3.4 points on the autonomy score and 2.7 on the safety score. Pay is reported per $10 an hour, in 2017 dollars.[8] The effects of the quality scores, occupation pay and the outside options are judged with standard errors clustered on occupation and those of the policy measures with standard errors clustered on state. Coefficients for every variable are in Appendix B, and Online Appendix Tables OA1 to OA10 report them with standard errors clustered on state and on occupation.

[8] The interquartile ranges are computed on the occupation-change sample and applied to every model, including the group estimates, so that effects are comparable across them. They are 0.09 for each half of the occupation pay deviation, $9.63 an hour for the pay gap, and 0.05 for both the pay deviation gain and the switcher wage premium. The figures report effects across the interquartile range of each measure, with occupation pay and the pay gap per $10 an hour. Where the text compares the destinations with the occupation held, the destinations are moved by the interquartile range of the destination level, 1.73 for autonomy and 1.06 for safety.

A. Employment at Both Dates

1. The Main Model

We find that job quality has modest effects on whether a worker was employed at both dates. An interquartile move up the autonomy score raises the probability of being employed a year later by 0.76 percentage points, an average that, as the period model shows, comes from the pandemic period. The same move on the safety score lowers it by 0.36 percentage points, so the two dimensions of job quality pull in opposite directions on this margin. A possible explanation for the negative effect of the safety score is that the occupations with the least contact with others are among the most physically demanding, and physically demanding work is associated with earlier exit from the labor force (Filer and Petri 1988; Hayward et al. 1989). Each $10 an hour of occupation pay raises the probability of being employed a year later by 1.56 percentage points, and pay growth in the occupation since 2016-17 has no significant effect.[9]

We find that the household matters for the employment decision. An employed spouse or partner raises the probability of being employed a year later by 5.0 percentage points, counting the spouse's employment and the characteristics of the spouse's job together, and a spouse in a better-paid occupation lowers it slightly.[10] Women are 2.75 percentage points less likely than men to be employed a year later. A child under six lowers the probability by less than half a percentage point, a small average that combines opposite effects for women and men (Section V.A.3).

[9] With standard errors clustered on occupation, the p values are 0.02 for the autonomy score, 0.03 for the safety score and below 0.001 for occupation pay. Online Appendix Table OA1 reports the coefficients of the main model with standard errors clustered on state and on occupation, and Tables OA2 to OA10 do the same for the other models.

[10] The effect compares partnered workers whose spouse is employed, with the same workers had the spouse not been employed, with every spouse variable set to its average among partnered workers whose spouse is not employed. By comparison, the average marginal effect of spouse employment on its own is 6.68 percentage points.

We find that the timing of the policy measures matters for employment at both dates (see Figure 2). A stay-at-home order in force during the six months before the second observed date lowers the probability of being employed a year later by 0.86 percentage points, while one in force during the six months before that raises it by 1.27 percentage points, a pattern consistent with job losses while an order was in force followed by recovery after it lifted. A mask mandate in the recent six months lowers the probability by 0.97 percentage points, and Extended Benefits in the earlier six months raise it by 0.99 percentage points. Because the effects of the two halves offset, no dated measure has a significant effect over a full year in force. School closure is the exception, as moving from no local schools closed to all of them closed for the school year lowers the probability by 1.69 percentage points. The small full-year effects are in line with Lin and Meissner (2020) and Chetty et al. (2020). Gupta et al. (2023) and Beland, Brodeur, and Wright (2020) find that stay-at-home orders reduced employment in the first months of the pandemic. Because our measure compares employment at two dates a year apart, a worker who lost a job while an order was in force and was reemployed before the second date is counted as employed at both, and the negative effect of an order in force during the recent six months is likely the part of those losses that our measure records.

Relative to high school graduates, workers without a high school diploma are 2.03 percentage points less likely to be employed a year later and those with an advanced degree are 2.02 percentage points more likely, with some college and four-year degrees falling in between. Relative to white non-Hispanic workers, Black non-Hispanic workers are 2.17 percentage points less likely to be employed at both dates, other non-Hispanic workers are 0.80 percentage points less likely, and there is no significant difference for Hispanic workers. Employment at both dates does not differ between metropolitan and non-metropolitan areas. Workers in public

administration and finance are about 2.0 and 1.5 percentage points more likely to be employed at both dates than those in mining, the omitted industry, and no other industry differs significantly (Appendix Figure A1).[11]

2. Differences across Pandemic Periods

During the pandemic, our results suggest that autonomy raised the probability of being employed a year later. An interquartile move up the score increases it by 1.31 percentage points from March 2020 through 2022, with no significant effect before or after (see Figure 3).[12] Within the pandemic period the effect is concentrated in 2020, at 2.59 percentage points from March to December 2020 against 0.66 in 2021 to 2022.[13] This is consistent with work that could be paused, paced and done on a predictable schedule being better able to absorb the pandemic shock. The autonomy score is also highly correlated with the ability to work from home. Across occupations, weighted by employment, its correlation with the ORS share of workers whose job can be done remotely is 0.76, against 0.07 for the safety score, so the pandemic-period effect of autonomy likely overlaps the work-from-home channel documented by Dingel and Neiman (2020) and Mongey, Pilossoph, and Weinberg (2021). The safety score lowers the probability of being employed a year later before and after the pandemic, and its effect during the pandemic is

[11] Industry pay is constant within broad industry, so each industry effect is computed with the industry indicator and the industry pay level set together at that industry's values.

[12] With standard errors clustered on occupation, the effect of autonomy has p values of 0.55, 0.002 and 0.27 across the three periods. It is larger during the pandemic than before it, by 1.49 percentage points across the interquartile range of the score ($p < 0.001$), and larger than after it ($p = 0.02$).

[13] These estimates are from the model with the pandemic period split in two, with standard errors clustered on state; that model was not estimated with standard errors clustered on occupation. Before and after the pandemic the effect of autonomy is not distinguishable from zero. The effect of the safety score is not distinguishable from zero in 2020 and lowers the probability of being employed a year later by 0.41 percentage points in 2021 to 2022. The boundary between 2020 and 2021 to 2022 coincides with the change in policy regime, since mask mandates, stay-at-home orders and business closures are concentrated in 2020, so a difference between the two halves could reflect the policies in force as well as how workers responded.

of similar size but only marginally significant. Because the scores are fixed, a change in how exposed work was carried out during the pandemic would not show up in them.[14]

The effect of autonomy on employment at both dates also differs with the pay of the occupation and with school closure. It is concentrated in lower-paid occupations, with an interquartile move up the autonomy score raising the probability of being employed a year later by 0.85 percentage points at the 25th percentile of occupation pay and having no effect at the 75th. It is also smaller the higher an occupation's safety score (Online Appendix Table OA2). Over the 2020-21 and 2021-22 school years, an interquartile move up the autonomy score raises the probability of being employed a year later by 2.53 percentage points where all local schools were closed for the school year and has no significant effect where none were, a difference of 2.11 percentage points. School closure also changes the effect of the safety score. Its negative effect, 0.75 percentage points across its interquartile range where no local schools were closed, is not distinguishable from zero where all were. No other policy measure changes the effect of either score significantly (Appendix Tables B4 and B5).[15]

Occupation pay raises the probability of being employed a year later in every period, by about 2 percentage points per $10 an hour. The pay deviations, not significant before the pandemic, are positive during it, significantly so for the distant half, but the effects are small, about 0.2 percentage points across their interquartile ranges (see Figure 4).[16]

[14] With standard errors clustered on occupation, the effect of the safety score has p values of 0.001, 0.07 and below 0.001 across the three periods, and it does not change significantly across them.
[15] With standard errors clustered on occupation, the effect of autonomy at the 25th percentile of occupation pay has a p value of 0.006 and the difference between the 75th and 25th percentiles 0.005. With school closure the difference has a p value of 0.009 for autonomy and 0.03 for safety, and no other policy measure changes the effect of either score with a p value below 0.10. The coefficient on the interaction of the two scores has a p value below 0.001.
[16] With standard errors clustered on occupation, occupation pay has p values below 0.001 in every period. During the pandemic the recent pay deviation has a p value of 0.06 and the distant one 0.008.

The pay movements ran against quality. Within each month, the correlation between an occupation's recent pay deviation and its autonomy score fell from -0.16 before the pandemic to -0.33 during it and -0.40 after, so the occupations with the least autonomy were those whose pay rose most. The safety score shows no such pattern (see Figure 5). Because low-autonomy occupations are also among the lowest paid, the pattern for autonomy is consistent with the wage compression documented by Autor, Dube, and McGrew (2023), which raised pay most at the bottom of the distribution, although the correlation does not separate autonomy from the level of pay.[17]

The same model lets the education and race differences vary across the three periods (see Figure A2). The movement in education is concentrated in advanced degrees as the protection an advanced degree gave against losing employment rose from 1.41 percentage points before the pandemic to 2.75 during it, a significant increase, and settled at 1.95 percentage points after. The other education groups barely move relative to high school graduates. The employment gap relative to white non-Hispanic workers did not change significantly across the periods.

3. Gender Differences

The third model interacts gender with every variable, including the state and year-month effects, so that women's and men's effects are equivalent to estimating them separately and every difference between them can be tested. The employment equation covers 764,576 records for women and 836,803 for men and the occupation equation the 682,078 and 766,008 records employed at both dates. Within each, the presence and ages of children are interacted with the policy measures, the quality scores and the outside options.

[17] For the distant half of the pay deviation the correlations are -0.13, -0.28 and -0.46. With the safety score they lie between 0.02 and 0.06 in every period.

We report two versions. The first splits children into three age bands, under six, six to twelve and thirteen to eighteen. The second uses two bands, under six and six to eighteen, and adds interactions of children with period, of the outside options with period, and of all three together. The gender estimates they produce agree to the third decimal, in percentage points, everywhere except the two outside-option pay measures that the second version interacts with period, discussed in Section V.B.3.

A child under six lowers the probability of being employed a year later by 3.30 percentage points for women and raises it by 1.62 percentage points for men, the direction Pitts (2021) finds for women with young children in 2020, although our estimate covers 2018 to 2025 and holds the policy measures, including school closure, fixed.[18] Occupation pay raises the probability of being employed a year later by more for women than for men, 1.76 against 1.30 percentage points per $10 an hour. The effects of the two quality scores do not differ significantly between them (see Figure 6). The effect of autonomy is significant and positive for both, and the negative effect of safety is significant for men but not for women.[19]

The effect of school closure on employment at both dates differs by the ages of the children in the household (see Figure 7). For women there is no significant effect where there is a child under six, while closure lowers the probability of being employed a year later by 3.78 percentage points where there is a child six to twelve; the effect where there is a teenager is similar in size but not significant at the 5 percent level. For men, closure lowers the probability where there is a child six to twelve and where there is a teenager.[20] This suggests that school closures operated

[18] The difference between women and men is significant for a child in every band ($p < 0.001$; Appendix Table B3), at 4.92 percentage points for a child under six, 1.49 for a child six to twelve and 0.68 for a child thirteen to eighteen.
[19] With standard errors clustered on occupation, the p values for the autonomy score are 0.02 for women and 0.02 for men, for the safety score 0.11 and 0.04, and for occupation pay below 0.001 for both.
[20] For men, closure lowers the probability of being employed a year later by 2.37 percentage points where there is a child six to twelve ($p = 0.025$) and by 2.74 where there is a teenager ($p = 0.04$), and has no significant effect where there is a child under six.

differently than care for younger children, and agrees with Landivar et al. (2023) on child age, who find employment losses among mothers of children five to twelve substantially larger than among mothers of preschoolers and attribute them to school closures lasting longer than closures of childcare (see also Collins et al. 2021; Landivar et al. 2022). Note that their comparison is by the age of the youngest child, while ours is by the presence of a child in each band, so the two are not measured identically. Where there is a child six to twelve, the point estimate for women is larger than for men, but the difference is not statistically significant, and the estimates are imprecise enough that the gap could be as large as 4 percentage points.[21]

4. Heterogeneity

This subsection re-estimates the main model separately within groups, so that nothing is constrained to be the same across them. The groups are three education groups (high school or less, some college, a bachelor's degree or more), five age groups (18 to 21, 22 to 24, 25 to 54, 55 to 64, and 65 and over), thirds of the occupation pay distribution, and four industry groups chosen because the pandemic reached them very differently (education, health care, leisure and hospitality, and all other industries). Effects are scaled by the same interquartile ranges as the main model, not by ranges within each group.

The effect of autonomy on being employed at both dates rises with education and with age (see Figure 8). It is significant for workers with a bachelor's degree or more and for every age group from 25 on, but not for the two lower education groups, the two youngest age groups or any third of occupation pay. The point estimate for the youngest workers is negative, which may reflect school enrollment or early-career mobility. The negative effect of the safety score is largest

[21] The child bands are not mutually exclusive. A worker with children in more than one band appears in each, so every effect reported by child band is for workers with a child in that band, not a comparison with households without children.

among workers 65 and over and is also significant for workers with a bachelor's degree or more, workers 25 to 54 and the middle and best-paid thirds of occupation pay.[22]

Occupation pay runs the other way. Its effect on being employed at both dates falls with education, and it is far larger in the lowest third of the occupation pay distribution than in the other two. Each additional $10 an hour raises the probability of being employed a year later by 4.9 percentage points in the lowest third and 0.5 percentage points in the highest and has no significant effect in the middle third (see Figure 9). Where the worker's own earning capacity is lowest, the partner's employment matters more, and a child under the age of six lowers the probability of being employed a year later by 3.05 percentage points among 22- to 24-year-olds and by 1.46 in the lowest pay third, against 0.45 for the sample as a whole.

Autonomy matters most for employment at both dates in leisure and hospitality and in health care, two of the sectors hit hardest by the pandemic, and is significant there and in the residual industry group but not in education (see Figure 8). The safety score has its largest negative point estimate in leisure and hospitality, but it is not significant in any of the four industry groups. An employed spouse matters most in leisure and hospitality.

B. Occupation Change

1. The Main Model

The outside-option quality measures are the average gain in each score among workers who have left the same occupation, so the model distinguishes the quality of the occupation a worker holds from the quality of those they could move to. Because the gain is destination minus origin,

[22] With standard errors clustered on occupation, the p values for the autonomy score are 0.20, 0.07 and below 0.001 across the three education groups; 0.13, 0.24, 0.004, below 0.001 and 0.003 across the five age groups; 0.58, 0.23 and 0.14 across the thirds of occupation pay; and 0.41, below 0.001, 0.02 and 0.006 across education, health care, leisure and hospitality and the residual group. For the safety score they are 0.14, 0.07 and 0.002; 0.32, 0.23, 0.04, 0.24 and 0.004; 0.28, 0.047 and below 0.001; and 0.76, 0.51, 0.17 and 0.23.

the estimated effect of the origin score holds the gain fixed, so we report the effect of the origin score holding the destinations fixed instead, which is that estimate less the effect of the gain.[23] An interquartile move in the autonomy of the destinations raises the probability of changing occupation by 7.4 percentage points. Holding the quality of the destinations fixed, the autonomy of the occupation a worker holds has a smaller point estimate, 3.4 percentage points for an interquartile move,[24] which is not significant, and the difference between the two is not significant either. For safety the pattern reverses. An interquartile move up the safety score of the occupation held raises the probability of changing occupation by 2.9 percentage points, while the effect of the safety of the destinations, 1.3 percentage points, is not significant.[25]

Our results therefore suggest that the autonomy of the work within reach draws workers out of their occupations, while the autonomy of the job they hold has a smaller effect that is less precisely estimated. The scores of the occupation held and the outside options are closely related, which limits how separately their effects can be identified. The gain is negatively correlated with the origin score by construction, at -0.91 for autonomy and -0.92 for safety, and the correlation between the origin score and the destination level is 0.84 for autonomy and 0.77 for safety. Because the outside-option measure is built from where leavers actually went, it is determined jointly with how many leave and can also reflect occupational ladders.

[23] The estimated effect of the origin score itself, which holds the gain fixed, is 17.9 percentage points across the interquartile range of the autonomy score and 6.2 across that of the safety score. Moving the destinations by the interquartile range of the origin score instead, their effects are 14.6 and 3.2 percentage points.

[24] The positive point estimate for the autonomy of the occupation held reflects moves between management occupations, which are coded finely. Counting moves from one management occupation to another as no change lowers it to 0.07 percentage points, and dropping moves into management or supervisory occupations lowers it to 1.06.

[25] With standard errors clustered on occupation, the p values are 0.008 for the autonomy of the destinations, 0.12 for the autonomy of the occupation held and 0.37 for the difference between them, and 0.02 for the safety of the occupation held and 0.35 for the safety of the destinations. The coefficients from which these effects are computed are in online Appendix Table OA1.

Pay appears to hold workers in their occupations. Holding the pay of the destinations fixed, each $10 an hour of occupation pay lowers the probability of changing occupation by 3.9 percentage points, the reverse of its effect on employment at both dates. Recent pay growth in the occupation has a small positive effect that is only marginally significant. None of the three outside options for pay is significant. The point estimates suggest that a larger pay gap to the destination occupations deters movement and that the pay deviation gain and the switcher wage premium encourage it, but the effects are small.[26]

The positive effect of an occupation's own safety partly reflects where workers can go from it, since the origin safety score and the safety gain are strongly negatively correlated (-0.92). Holding the safety of the destinations fixed lowers the effect across the interquartile range of the score from 6.2 to 2.9 percentage points, so our results suggest that about half of it reflects where workers can go.

The household has small effects on the occupation decision. A child under six lowers the probability of changing occupation by 1.22 percentage points, while a child of thirteen to eighteen raises it. The gender gap is smaller than on the employment margin. A business closure in force during the six months before the second date raises the probability of changing occupation by 1.51 percentage points; no other half-year effect is significant, and no measure has a significant effect over a full year in force.

Workers with some college are the most likely to change occupation and those with an advanced degree the least likely. Black non-Hispanic workers are 4.70 percentage points more likely than white non-Hispanic workers to change occupation, so Black workers are both less

[26] With standard errors clustered on occupation, the p values are 0.01 for occupation pay, 0.08 for recent pay growth, 0.32 for the pay gap, 0.17 for the pay deviation gain and 0.31 for the switcher wage premium. Across their interquartile ranges the pay gap lowers the probability by 2.49 percentage points and the pay deviation gain and the switcher wage premium raise it by 0.54 and 0.29.

likely than white workers to be employed at both dates and more likely to change occupation. Living outside a metropolitan area lowers the probability of changing occupation by 1.61 percentage points, consistent with a thinner local labor market offering fewer options. Relative to mining, workers in most industries are between 8 and 16 percentage points less likely to change occupation; manufacturing, trade and information do not differ significantly from it (Appendix Figure A1).

2. Differences across Pandemic Periods

We find no change across the periods in the effect of job quality on occupation change. Holding the destinations fixed, the point estimate of the effect of an occupation's own autonomy dips during the pandemic and returns after it, that of its own safety rises slightly after 2022, and the effects of the autonomy gain and the safety gain available elsewhere fall after 2022 (see Figures 3 and 10), but none of these changes is significant.[27] The absence of a change is consistent with the little excess reallocation across jobs that Forsythe et al. (2022) find. The autonomy of the destinations raises the probability of changing occupation in every period. The autonomy of the occupation held raises it before and after the pandemic but not during it, and the safety of the destinations in no period.[28]

Holding the pay of the destinations fixed, occupation pay holds workers in place by about 4.2 percentage points per $10 an hour in every period (see Figure 4). The autonomy and safety scores do not interact on this margin. A stay-at-home order changes the effect of the autonomy of

[27] With standard errors clustered on occupation, the changes have p values of 0.09 for the autonomy of the occupation held during the pandemic and 0.80 after it, 0.39 for its safety after 2022, and 0.28 and 0.055 for the autonomy and safety gains after 2022. Under state clustering all of them but the change in own autonomy after 2022 would be significant. The underlying effects are, across the interquartile range of each score and holding the destinations fixed, 4.22, 2.97 and 4.47 percentage points for autonomy and 3.27, 2.76 and 3.78 for safety in the three periods.

[28] With standard errors clustered on occupation, the p values for the autonomy of the destinations are 0.007, 0.01 and 0.04 across the three periods, for the autonomy of the occupation held, holding the destinations fixed, 0.046, 0.23 and 0.046, and for the safety of the destinations 0.37, 0.38 and 0.95.

the occupation held, holding the destinations fixed, which is 4.92 percentage points larger across its interquartile range when an order is in force for a full year. No other policy measure changes the effect of either score on occupation change.[29] Holding the outside options at their pre-2020 values for each origin occupation, as in equation 3, leaves the effects of the quality and pay of the occupation held and of the autonomy of the destinations close to their estimates with the rolling measures.[30]

What the alternatives pay moved more across periods, but its effects remained small. The pay deviation gain raised the probability of changing occupation by 0.86 percentage points across its interquartile range during the pandemic, the only period in which it is significant. The point estimates of the switcher wage premium rise from negative before the pandemic to positive after it, but it is not significant in any period.[31] The rise in the point estimates is consistent with the gap in wage growth between job switchers and job stayers in the Atlanta Fed's Wage Growth Tracker, which widened during the pandemic, reaching a series high of 2.7 percentage points in November 2022, and remained above its pre-pandemic level afterward. It is also consistent with the wage gains for job changers documented by Hotchkiss (2021) and Autor, Dube, and McGrew (2023).

[29] The p value of the difference is 0.010 with standard errors clustered on occupation and 0.002 with standard errors clustered on state. No other policy measure changes the effect of either score with a p value below 0.05 under occupation clustering.

[30] Online Appendix Tables OA2 and OA10 report the period model with the rolling and the fixed outside options, respectively. With the fixed measures, the effect of the autonomy of the destinations is 6.8 percentage points across its interquartile range, against 7.0 with the rolling measures, and the coefficients on the quality and pay of the occupation held change little. The fixed measures drop the 168 records, 156 of them employed at both dates, whose origin occupation has no records before March 2020.

[31] With standard errors clustered on occupation, the pay deviation gain has p values of 0.70, 0.04 and 0.48 across the three periods and the switcher wage premium 0.39, 0.59 and 0.32. Across its interquartile range the switcher wage premium runs -0.29, 0.21 and 0.51 percentage points across the periods. With the outside options fixed at their pre-2020 values the pay deviation gain is no longer significant during the pandemic and the switcher wage premium changes sign, at -0.42 percentage points averaged across the three periods of the period model against 0.12 with the rolling measures.

A single pandemic period from March 2020 to December 2022 spans both the initial shock and the reopening and quits of 2021 and 2022, so we also estimated the model with that period split into the rest of 2020 and 2021 to 2022. The point estimates of the responses to the autonomy of the occupation held and to the autonomy of the destinations are similar in the two halves, which suggests that the three-period results above are not an average of two different pandemic responses. The two pay measures do differ between the halves, but not as a step. The switcher wage premium builds gradually across the four periods and the pay deviation gain peaks in 2021 to 2022, and the 2020 estimates of both are not distinguishable from zero. We treat the comparison between the halves as descriptive rather than as a test of whether behavior changed. The boundary between them coincides with the policy regime, since mask mandates, stay-at-home orders and business closures are concentrated in 2020 while vaccine mandates and Extended Benefits run into 2021 and 2022, so a difference between the halves could reflect workers responding differently or the policies in force differing. A record observed in 2020 also spans a window that begins before the shock, while one observed in 2022 has had two years to respond. Finally, records after 2020 condition on having remained employed through it, where records in 2020 condition on pre-pandemic employment.[32]

3. Gender Differences

Using the same gender model, on the occupation margin the safety and pay of the occupation a worker holds are significant for men but not for women, and its autonomy is significant for neither (see Figure 6). Holding the destinations fixed, the point estimates for an interquartile

[32] Holding the destinations fixed, the effect of the autonomy of the occupation held is 2.63 percentage points in the rest of 2020 and 3.16 in 2021 to 2022, against 4.22 before the pandemic, and that of the autonomy of the destinations 6.65 and 7.34 against 7.63. Across their interquartile ranges the switcher wage premium runs -0.29, 0.11, 0.27 and 0.51 percentage points across the four periods and the pay deviation gain 0.16, 0.58, 0.89 and 0.39, with the 2020 estimate of each not distinguishable from zero. These estimates use standard errors clustered on state.

move up the autonomy score are 1.3 percentage points for women and 4.8 for men, but neither is significant and neither is the difference between them. The safety score raises the probability of changing occupation by about 3 percentage points for both, significantly for men. Each $10 an hour of occupation pay lowers it by 3.0 percentage points for women and 4.7 for men, again significantly only for men.[33] Women work in almost as many occupations as men but are more concentrated within them, so clustering on occupation leaves fewer effective groups for women and their estimates are less precise.[34]

As in the main model, the autonomy of the destinations has the larger point estimate. An interquartile move in the autonomy of the destinations raises the probability of changing occupation by 8.8 percentage points for men and 6.0 for women, significantly for men and marginally for women. For safety, the occupation held has the larger effect for both, 2.9 against 1.0 percentage points for women and 3.1 against 1.7 for men, and the safety of the destinations is not significant for either.[35]

The outside options for pay matter little for either sex. The pay deviation gain draws men out of their occupations, by 0.83 percentage points across its interquartile range, and does nothing

[33] With standard errors clustered on occupation, holding the destinations fixed, the p values for women are 0.57 for the autonomy of the occupation held, 0.08 for its safety and 0.09 for its pay, and for men 0.06, 0.02 and 0.004. The women-minus-men difference in the effect of the autonomy of the occupation held has a p value of 0.30, treating the two fits as independent, against below 0.001 under state clustering.

[34] Women appear in 468 occupations and men in 472, but women's records are more concentrated, giving an effective 78.3 occupation clusters for women against 94.9 for men. Online Appendix Tables OA3 to OA5 report the coefficients of the gender models.

[35] The women-minus-men differences, from the pooled gender model with standard errors clustered on state, holding the destinations fixed, have p values below 0.001 for the autonomy score and occupation pay and of 0.55 for the safety score. For the outside options they are below 0.01 for the autonomy gain, the safety gain, the pay gap and the pay deviation gain, and 0.06 for the switcher wage premium. On employment at both dates they are below 0.001 for occupation pay and 0.12 and 0.33 for the autonomy and safety scores. Moving the destinations by the interquartile range of the origin score instead, the autonomy of the destinations raises the probability of changing occupation by 11.8 percentage points for women and 17.5 for men. These differences have not been tested with standard errors clustered on occupation, except for the autonomy of the occupation held.

measurable for women, while the pay gap and the switcher wage premium are not significant for either (see Figure 11).[36]

How strongly a worker responds to the outside options also depends on the ages of the children in the household (see Figure 12). The autonomy of the destinations draws men out of their occupations in every child band, and there is no significant effect for women across the age bands. For men the pay deviation gain is significant only where there is a child under six, and for women it is not significant in any band. The response to the pay gap differs most for women with a child under six, for whom a larger gap does not deter switching, and the difference from women without a child under six is significant, possibly because caring for a young child limits which moves are possible.[37] Testing the interactions as blocks, with standard errors clustered on state, shows that for both women and men the response to the outside options differs significantly by whether there is a child under six, and for no other age group, while the response to the quality of the occupation a worker holds does not differ by the ages of their children at all.[38] The response to the policy measures differs sharply by the presence of young children for men and much less so for women.[39]

[36] With standard errors clustered on occupation, the p values for the autonomy of the destinations are 0.08 for women and 0.001 for men, for the safety of the destinations 0.58 and 0.15, for the pay gap 0.67 and 0.16, for the pay deviation gain 0.77 and 0.03, and for the switcher wage premium 0.30 and 0.55.

[37] With standard errors clustered on occupation, for women the autonomy of the destinations has p values of 0.22, 0.11 and 0.10 with a child under six, six to twelve and thirteen to eighteen, and for men below 0.001 in all three. For men the pay deviation gain has p values of 0.02, 0.12 and 0.10. For women with a child under six the effect of the pay gap has a p value of 0.59 and its difference from women without a child under six a p value below 0.001; for men the same difference has a p value of 0.004.

[38] The block of interactions between the outside options and the presence of a child under six is significant for both genders, with a chi-squared statistic of 52.4 for women and 14.6 for men, on five degrees of freedom. For the two older child bands among women the statistics are 3.2 and 2.0, neither significant. The corresponding block for the two quality scores of the origin occupation is not significant for either gender or any band. These block tests use standard errors clustered on state.

[39] The block of interactions between the policy measures and the presence of a child is far larger for men than for women. On twelve degrees of freedom the statistics are 100.8 for men with a child under six and 62.1 with a child six to twelve, against 16.3 and 29.9 for women. The women's block for a child under six does not reach conventional significance, although individual measures within it do, including school closure.

The second version of the model, with the three periods, shows that the change in how strongly the outside options draw workers out of their occupations is concentrated in households with a child under six.[40] The two versions agree except on the two outside-option pay measures that the second interacts with period, and estimating the coarser child grouping without the period dimension shows that this difference comes from adding period rather than from grouping the child ages more coarsely.[41]

4. Heterogeneity

Figure 13 reports the quality of the occupation held by group and Figure 14 its pay, with Figures 15 and 16 the outside options for quality and pay. Using the same education, age, occupation-pay and industry groups as for employment at both dates, and holding the destinations fixed, the point estimates of the effect of the autonomy of the occupation a worker holds vary considerably across groups, from negative in the lowest-paid third of occupations, in education and in leisure and hospitality to 9.1 percentage points across the interquartile range of the score in health care (see Figure 13). The effect is significant only for workers aged 22 to 24.[42]

The autonomy of the destinations is significant for workers with some college or a bachelor's degree, for every age group from 25 on, for the best-paid third of occupations, and in education and the residual industry group (see Figure 15). Its point estimate exceeds that of the autonomy

[40] The three-way block of children, outside options and period is significant for both genders, with chi-squared statistics of 44.4 for women and 58.0 for men on twenty degrees of freedom. Taking the child bands one at a time, on ten degrees of freedom, only the youngest band clears, at 24.9 for women with a child under six against 8.8 for a child six to eighteen, and 20.4 against 14.8 for men. These block tests use standard errors clustered on state.

[41] Across its interquartile range, the pay deviation gain raises the probability of changing occupation by 0.14 percentage points for women and 0.83 for men in the three-band version, against 0.07 and 0.77 in the period version. The switcher wage premium raises it by 0.37 and 0.16 against 0.49 and 0.29.

[42] With standard errors clustered on occupation, the p values for the autonomy of the occupation held, holding the destinations fixed, are 0.29, 0.13 and 0.81 across the three education groups; 0.30, 0.008, 0.09, 0.27 and 0.85 across the five age groups; 0.80, 0.20 and 0.29 across the thirds of occupation pay; and 0.30, 0.053, 0.12 and 0.11 across education, health care, leisure and hospitality and the residual group. Every group has between 131 and 472 occupation clusters. Online Appendix Tables OA6 to OA9 report the coefficients for the education, age, occupation pay and industry groups.

of the occupation held in most groups, with the largest gap in the best-paid third, at 13.2 against 4.2 percentage points across the interquartile range of each, but the gap is significant only in education and in leisure and hospitality.[43]

The safety of the occupation held raises the probability of changing occupation significantly for workers with some college or a bachelor's degree, workers 55 and over, the middle third of occupation pay, and in education and health care, where its effect is largest, at 8.7 percentage points. The safety of the destinations matters more than the safety of the occupation held in the lowest and best-paid thirds of occupation pay and in leisure and hospitality and the residual industry group, while in education and health care its point estimates are negative but not significant.[44] Holding the pay of the destinations fixed, occupation pay lowers the probability of changing occupation significantly for workers with a bachelor's degree or more, workers 22 to 54 and the best-paid third, and in health care and the residual group, and its positive point estimates for workers with a high school education or less, in education and in leisure and hospitality are not significant. Among the outside options for pay, only the pay deviation gain is significant in any group, for the youngest workers and the residual industry group.[45]

[43] With standard errors clustered on occupation, the p values for the autonomy of the destinations are 0.14, 0.004 and 0.04 across the three education groups; 0.71, 0.054, 0.008, 0.01 and 0.02 across the five age groups; 0.45, 0.68 and 0.02 across the thirds of occupation pay; and 0.008, 0.25, 0.10 and 0.002 across the four industry groups. For the difference between the two effects they are 0.85, 0.34 and 0.23; 0.83, 0.98, 0.39, 0.37 and 0.22; 0.54, 0.61 and 0.32; and 0.03, 0.79, 0.04 and 0.38. Moving the destinations by the interquartile range of the origin score instead, the gap in the best-paid third is 26.1 against 4.2 percentage points.

[44] With standard errors clustered on occupation, the p values for the safety of the occupation held, holding the destinations fixed, are 0.08, 0.046 and 0.04 by education; 0.53, 0.29, 0.06, 0.03 and 0.004 by age; 0.98, 0.03 and 0.54 by occupation pay; and 0.02, below 0.001, 0.56 and 0.14 by industry. For the safety of the destinations they are 0.38, 0.045 and 0.78; 0.17, 0.27, 0.29, 0.44 and 0.88; 0.008, 0.75 and 0.01; and 0.39, 0.58, 0.02 and 0.03.

[45] With standard errors clustered on occupation, the p values for occupation pay, holding the pay of the destinations fixed, are 0.18, 0.43 and 0.002 by education; 0.38, 0.03, 0.002, 0.09 and 0.52 by age; 0.40, 0.07 and 0.02 by occupation pay; and 0.41, below 0.001, 0.57 and 0.04 by industry. The pay deviation gain has p values of 0.003 for workers 18 to 21 and 0.004 in the residual industry group, and no other outside option for pay has a p value below 0.05 in any group.

Leisure and hospitality lost 49 percent of its employment in the first two months of the pandemic, yet the quality of the occupation its workers hold has no significant effect on whether they leave it. What matters for them is the safety of the occupations available elsewhere, which raises the probability of changing occupation by 3.8 percentage points for an interquartile increase in the safety of the destinations.[46] These estimates come from the smallest sample of the four industry groups, 94,839 records on the occupation margin against 980,767 in the residual group.

C. Did Workers Who Changed Occupation Move Toward Higher Quality Jobs?

Equation 3 asks whether, holding composition fixed, workers who changed occupation moved further toward autonomy or safety after March 2020 than before it. We treat the outside options in three ways. The rolling measures used in equation 2 are averages of where other workers leaving the same occupation went in the same year, which is close to an average of the outcome of equation 3 itself, so they can absorb a shift in the direction of moves, and we show that version for comparison only. We instead hold the outside options at their pre-2020 values for each origin occupation, which keeps the control for where workers from an occupation typically go without letting it move with the pandemic, and, as a check on how those values are constructed, we also omit the outside options entirely.

With the outside options at their pre-2020 values, we find that workers who changed occupation moved slightly further toward both autonomy and safety after March 2020 (see Figure 17). Their autonomy score rose by 0.07 points more during and after the pandemic than before it, and their safety score by 0.04 points more after 2022, while the gain in safety during

[46] Moving the destinations by the interquartile range of the origin score instead, the effect is 9.9 percentage points. These estimates cover the whole of 2018 to 2025 without separating the three periods, so they do not show whether the response in leisure and hospitality changed with the pandemic.

the pandemic is not significant. Omitting the outside options gives nearly the same result, although the gain in safety after 2022 is then only marginally significant. The gains are small, as 0.07 points on the autonomy score is about 2 percent of its interquartile range of 3.4 points.[47]

The timing is consistent with Fee (2022), who finds that mobility out of low-quality jobs contracted in 2020 and rebounded in 2021, and the small size of the gains is consistent with Forsythe et al. (2022), who find reallocation across jobs no greater than before the pandemic. Our results differ from those of Braakmann, Eberth, and Wildman (2022), who find that workers in the United Kingdom left high-risk occupations in 2020, whereas in our data the additional safety gained by workers who changed occupation appears only after 2022.

Switchers from the lowest-paid third of occupations gain close to a point of the autonomy score and those from the best-paid third lose close to a point, in every period, which likely reflects where the two groups start rather than any change over time. These are moves within a single year, measured from the occupation a worker held twelve months earlier, and by the later periods that starting point may already reflect earlier reallocation. The average change among switchers in the raw data rises to about 0.05 points of the autonomy score in 2022 and 2023 and returns to near zero in 2024 and 2025 (Online Appendix Figure OA8), so the raw gains appear to have been temporary, although the adjusted gain for the post period as a whole, which runs through March 2025, is as large as during the pandemic.

[47] With the outside options at their pre-2020 values, the adjusted change in the autonomy score is 0.070 points higher during the pandemic than before it (standard error 0.017) and 0.068 higher after it (0.014), and in the safety score 0.024 (0.016) and 0.040 (0.015). With the outside options omitted the differences are 0.064 (0.018) and 0.055 (0.016) for autonomy and 0.012 (0.016) and 0.030 (0.016, $p = 0.07$) for safety. With the rolling measures they are 0.025 and 0.022 for autonomy, the second significant, and -0.002 and 0.005 for safety. We do not use the rolling version because each rolling measure is nearly an average of the outcome of equation 3 itself, so controlling for it removes the shift being measured. The 68 switchers whose origin occupation has no pre-2020 records are omitted from the fixed version.

D. Robustness

Both outcomes compare two dates twelve months apart rather than the months between them, so a worker who lost a job and regained one within the window is recorded as employed at both dates. In the same way, a worker who changed occupation more than once within the year is recorded by the occupation held at the second date. An occupation change is recorded only when the four-digit code differs between the two dates, and the change also passes the validation screen. Of the 1,448,086 records employed at both dates, 49.6 percent have a raw code change, and the screen validates 59.4 percent of those.

We estimate the occupation equation under three definitions of the dependent variable, holding everything else identical (see Figure 18): the validated measure used throughout; a definition that keeps known changes and known non-changes and drops the changes the screen rejects; and the raw code comparison with no screen, the measure Hodge (2022) uses. Their base rates are 29.5, 36.9 and 49.6 percent. No sign and no conclusion changes across them. Under all three, the autonomy of the destinations raises the probability of changing occupation significantly and the autonomy of the occupation held, holding the destinations fixed, does not.[48] Because the base rates differ, the magnitudes are best compared relative to each. The effect of an occupation's own autonomy across the interquartile range of the score, holding the destinations fixed, is about 11 percent of the base rate under the validated measure, 14 percent when the rejected changes are dropped and 12 percent under the raw comparison. Misclassification of a binary outcome biases estimates toward zero only when it is unrelated to the regressors (Hausman, Abrevaya, and Scott-

[48] With standard errors clustered on occupation, the p values for the autonomy of the destinations are 0.008, 0.003 and below 0.001 under the validated definition, with the rejected changes dropped and under the raw comparison, and those for the autonomy of the occupation held, holding the destinations fixed, are 0.12, 0.11 and 0.10.

Morton 1998), so no definition is a guaranteed lower bound, and the check that matters is whether the screen treats occupations differently by their quality.

We find no evidence that the validation measure favors better occupations. The share of raw code changes the screen validates falls with the autonomy of the origin occupation, from 65 percent in the lowest decile to 55 percent in the highest (not shown), and does not vary with the safety score, so the positive effect of quality on occupation change is not produced by the screen. The screen also removes close to 40 percent of raw code changes in every period and for both genders. Nor does the change in occupational classification in January 2020 create spurious changes. Every match ending in 2020 pairs a 2010-coded and a 2018-coded record, so the January and February 2020 matches, which end before the pandemic, are the only test not confounded with it. The raw change rate for those matches is within 0.7 percentage points of the rate for matches ending in November and December 2019 and within 0.2 of that for January and February 2019. Dependent coding within each four-month block confirms a quarter of the changes the screen validates, as a month-to-month change of occupation code appears for 27.6 percent of them. Each twelve-month comparison contains three month-to-month transitions inside the interview blocks and nine inside the eight-month break, so if changes were spread evenly through the year about a quarter would appear within a block. That validated changes appear within a block at close to that rate is consistent with the screen retaining genuine changes timed as real changes would be.

Finally, a joint model that estimates the employment and occupation-change equations together, allowing their unobserved determinants to be correlated (a probit with sample selection, reported in Appendix C), gives quality-score effects within 0.07 percentage points across the interquartile range of the score of the main-model estimates.

VI. Conclusions and Policy Implications

This paper asks how the quality of an occupation, the autonomy it gives a worker and how exposed its work is, bore on two outcomes through the pandemic. The first is whether a worker was employed at both dates a year apart, and the second is whether a worker employed at both dates changed occupation.

Our results suggest that whether a worker was employed at both dates depended on autonomy only during the pandemic, and mainly in 2020, while the safety score lowered the probability of employment and occupation pay raised it. We also find that whether a worker changed occupation depended on the autonomy of the occupations within reach, on the safety of the occupation the worker held, and on pay, which held workers in their occupations. Workers in occupations with more autonomy were also somewhat more likely to change occupation, but this effect is smaller and less precisely estimated, and our results suggest it partly reflects moves between management occupations.

The pandemic therefore appears to have changed which workers were employed at both dates more than how workers weighed the quality of their jobs in deciding whether to change occupation. We find no change across the periods in the effect of job quality on occupation change. Workers who changed occupation gained slightly more autonomy after March 2020 and slightly more safety after 2022 than before, but the gains are small. The pay available from moving drew workers out of their occupations only during the pandemic, through the pay deviation gain, and its effect was small. For employers, our results suggest that the autonomy available elsewhere bears on whether workers leave, so retaining workers may depend on how the job compares with the alternatives within reach as well as on the features of the job itself.

The effects of the occupation held on whether a worker changes occupation are estimated more precisely for men than for women, because women's employment is more concentrated within occupations, and we cannot distinguish the responses of the two to autonomy. Our results also suggest that school closure reduced employment at both dates where there was a child six to twelve, for both women and men.

The pandemic altered a great many behaviors, both from a public health perspective and in the workplace. Job quality shaped whether workers changed occupation throughout the period, through the autonomy of the occupations within reach and the safety of the occupation they held, but we find no evidence that the pandemic changed the weight workers placed on it. The pandemic made exposed work riskier, yet the safety of the occupations within reach does not significantly draw workers out of their occupations in any period, and the additional safety gained by workers who changed occupation after 2022 was small.

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Table 1. Sample Means, by Period.

	All	Pre	During	Post
Outcomes (Percent)				
Employed at Both Dates	90.4	91.3	89.1	91.2
Changed Occupation, Among Those Employed at Both Dates	29.5	29.0	29.8	29.7
Occupation Held Twelve Months Earlier				
Autonomy Score	0.308	0.249	0.329	0.352
	(1.942)	(1.943)	(1.939)	(1.943)
Safety Score	-0.031	-0.041	-0.027	-0.023
	(1.813)	(1.823)	(1.810)	(1.806)
Occupation Pay (2017 Dollars an Hour)	28.19	27.77	28.33	28.53
	(10.74)	(10.63)	(10.74)	(10.84)
Occupation Pay Deviation, Recent Half	0.032	0.012	0.039	0.048
	(0.089)	(0.077)	(0.087)	(0.101)
Occupation Pay Deviation, Distant Half	0.026	0.009	0.037	0.034
	(0.086)	(0.070)	(0.087)	(0.098)
Industry Pay (2017 Dollars an Hour)	27.00	26.88	27.07	27.05
	(5.268)	(5.329)	(5.224)	(5.245)
Industry Pay Deviation	0.051	0.026	0.063	0.067
	(0.041)	(0.020)	(0.033)	(0.053)
Own Industry Demand Shock	0.013	0.016	-0.001	0.029
	(0.050)	(0.012)	(0.075)	(0.026)
Outside Options, Among Those Employed at Both Dates				
Autonomy Gain	-0.023	-0.032	-0.023	-0.014
	(1.242)	(1.256)	(1.227)	(1.245)
Safety Gain	0.115	0.109	0.109	0.129
	(1.308)	(1.303)	(1.304)	(1.320)
Pay Gap (2017 Dollars an Hour)	-0.459	-0.410	-0.430	-0.555
	(6.246)	(6.136)	(6.207)	(6.425)
Pay Deviation Gain	0.001	0.001	0.001	0.002
	(0.054)	(0.043)	(0.053)	(0.065)
Switcher Wage Premium	0.008	0.014	0.011	-0.005
	(0.063)	(0.071)	(0.061)	(0.052)
State Policies and Mandates (Share of Days in Force)				
Mask Mandate, Recent Half	0.093	0.000	0.250	0.000
	(0.257)	(0.000)	(0.373)	(0.000)
Mask Mandate, Distant Half	0.086	0.000	0.231	0.000
	(0.247)	(0.000)	(0.363)	(0.000)
Stay-at-Home Order, Recent Half	0.027	0.000	0.073	0.000

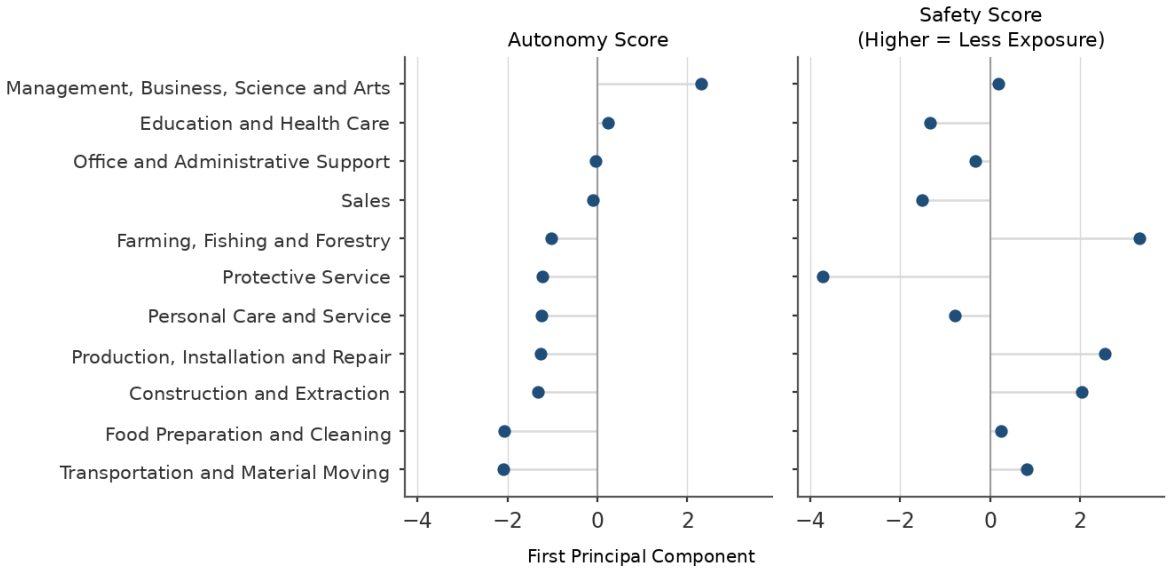
	All	Pre	During	Post
	(0.116)	(0.000)	(0.181)	(0.000)
Stay-at-Home Order, Distant Half	0.024	0.000	0.064	0.000
	(0.105)	(0.000)	(0.165)	(0.000)
Business Closure, Recent Half	0.072	0.000	0.195	0.000
	(0.209)	(0.000)	(0.307)	(0.000)
Business Closure, Distant Half	0.065	0.000	0.176	0.000
	(0.197)	(0.000)	(0.292)	(0.000)
Vaccine Mandate, Recent Half	0.047	0.000	0.118	0.013
	(0.190)	(0.000)	(0.294)	(0.069)
Vaccine Mandate, Distant Half	0.047	0.000	0.063	0.085
	(0.191)	(0.000)	(0.215)	(0.252)
Extended Unemployment Benefits, Recent Half	0.111	0.000	0.300	0.000
	(0.276)	(0.000)	(0.386)	(0.000)
Extended Unemployment Benefits, Distant Half	0.099	0.000	0.262	0.005
	(0.260)	(0.000)	(0.370)	(0.058)
School Closure Rate	0.062	0.000	0.166	0.000
	(0.153)	(0.000)	(0.214)	(0.000)
Household				
Married or Partnered (Percent)	65.1	65.1	65.6	64.6
Spouse Employed (Percent)	49.2	49.5	48.7	49.6
Spouse Occupation Pay (2017 Dollars an Hour)	15.20	14.91	15.26	15.47
	(16.71)	(16.46)	(16.74)	(16.97)
Spouse Occupation Autonomy Score	0.314	0.281	0.317	0.349
	(1.366)	(1.362)	(1.366)	(1.369)
Spouse Occupation Safety Score	-0.062	-0.063	-0.060	-0.062
	(1.258)	(1.265)	(1.257)	(1.250)
Spouse Pay Deviation, Recent Half	0.013	0.004	0.017	0.020
	(0.065)	(0.056)	(0.064)	(0.073)
Spouse Pay Deviation, Distant Half	0.011	0.003	0.016	0.012
	(0.062)	(0.051)	(0.064)	(0.070)
Spouse Industry Demand Shock	0.004	0.008	-0.001	0.007
	(0.037)	(0.012)	(0.057)	(0.019)
Spouse Industry Not Determined (Percent)	14.4	14.4	14.8	13.8
Spouse Has No Occupation (Percent)	14.2	14.3	14.5	13.7
Child Under 6 (Percent)	14.5	14.9	14.6	14.0
Child 6 to 12 (Percent)	20.2	20.6	20.1	19.9
Child 13 to 18 (Percent)	20.6	20.7	20.5	20.5

	All	Pre	During	Post
Demographics				
Woman (Percent)	47.7	47.9	47.6	47.7
Age (Years)	46.15	46.02	46.32	46.07
	(14.37)	(14.23)	(14.39)	(14.50)
Education (Percent)				
Less than High School	4.7	5.2	4.4	4.4
High School Graduate	26.1	26.5	25.9	25.9
Some College	27.6	28.6	27.5	26.4
Bachelor's Degree	25.5	24.7	25.7	26.3
Advanced Degree	16.1	15.0	16.4	17.0
Race and Ethnicity (Percent)				
White, non-Hispanic	70.6	71.3	71.1	68.9
Black, non-Hispanic	8.6	8.6	8.6	8.5
Other, non-Hispanic	7.7	7.5	7.6	8.2
Hispanic	13.1	12.7	12.7	14.3
Metropolitan Status (Percent)				
Metropolitan	80.9	80.6	80.8	81.5
Non-Metropolitan	18.1	18.4	18.3	17.5
Not Identified	1.0	0.9	0.9	1.0
Industry Twelve Months Earlier (Percent)				
Mining	0.6	0.7	0.7	0.5
Construction	7.0	6.9	7.1	7.0
Manufacturing	9.9	10.1	9.9	9.8
Trade	12.4	12.7	12.2	12.1
Transportation, Warehousing and Utilities	5.5	5.2	5.6	5.8
Information	1.8	1.8	1.7	1.7
Finance	6.9	6.8	7.1	6.9
Professional and Business Services	12.1	11.7	12.2	12.5
Education and Health	23.8	23.6	23.9	23.8
Leisure and Hospitality	7.4	7.9	7.1	7.4
Other Services	4.9	5.0	4.8	4.8
Public Administration	5.6	5.5	5.6	5.6
Agriculture	2.1	2.1	2.2	2.0
Sample Size				
Records	1,601,379	554,599	593,829	452,951
Employed at Both Dates	1,448,086	506,226	528,894	412,966

Notes: Means, with standard deviations in parentheses for continuous measures; indicators and categories are in percent. Pay is in 2017 dollars. Unweighted, over the matched records of the estimation sample; occupation change and the outside options are over records employed at both dates. The policy measures are the share of days in force in each six-month half of the twelve

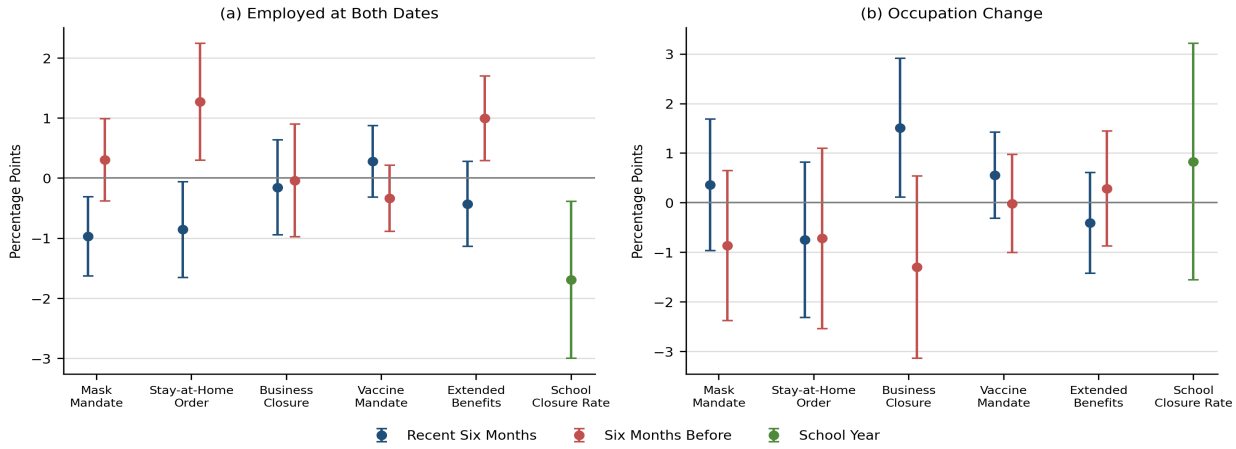
months before the second date, and the school closure rate is the share of local public schools that were closed. Pre is January 2018 through February 2020, during is March 2020 through December 2022, and post is January 2023 onward.

Figure 1. First Principal Component Combining ORS and O*NET Autonomy and Exposure Characteristics, by Broad Occupation.



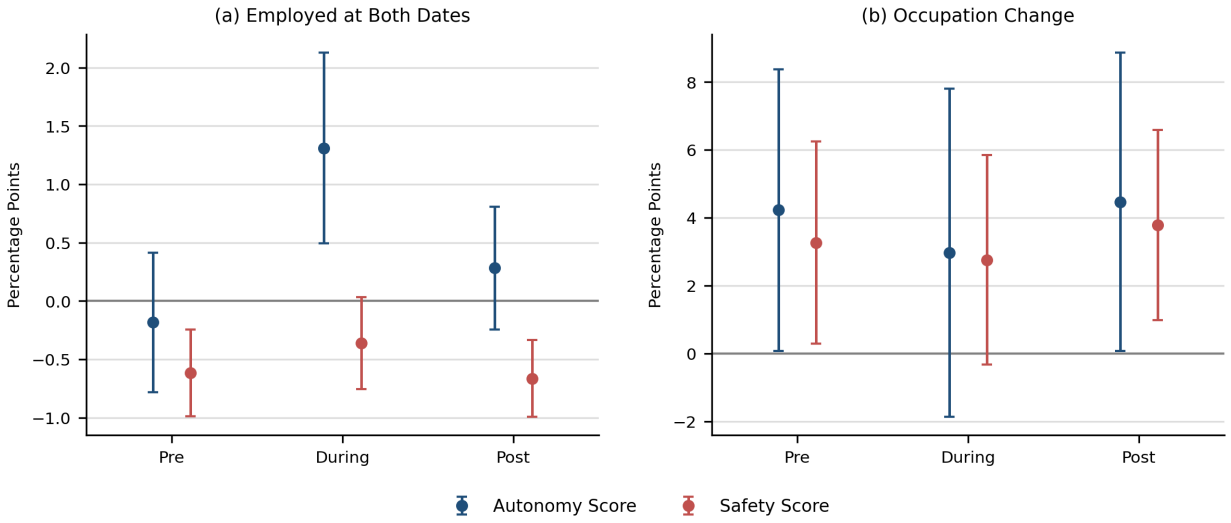
Notes: Occupations are ordered by the autonomy score. A higher safety score denotes less exposure. The occupation groups combine broad groups of the 2010 Census occupation classification, as set out in the notes to Online Appendix Figure OA5.

Figure 2. Marginal Effects of the State Policy Measures.



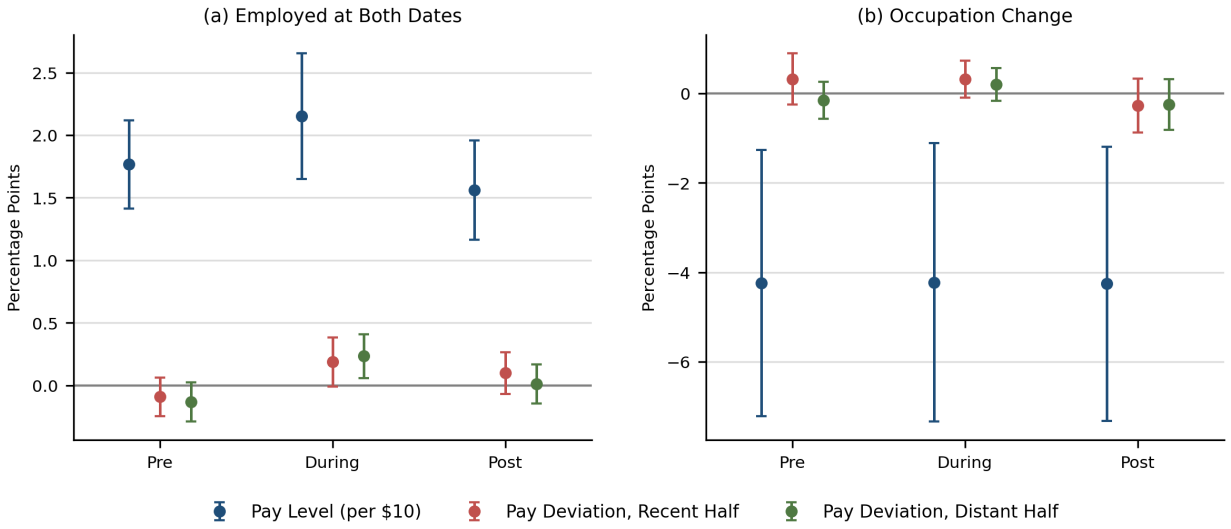
Notes: Average marginal effects from the main model, with ninety-five percent confidence intervals. For the mask mandate, stay-at-home order, business closure, vaccine mandate and Extended Benefits, each effect is that of the measure being in force for the whole of the six months before the second date (recent) or of the six months before that, per unit of the share of days in force. The school closure rate is the share of local public schools that were closed, from Parolin and Lee (2021), who measure closure from mobile-phone foot traffic, for the school year in which the record falls, taken as the mean of its first- and fourth-quarter rates; the effect shown is that of moving from no local schools closed to all of them closed for the school year. Effects are in percentage points against base rates of 90.4 percent for employment at both dates and 29.5 percent for occupation change. The two panels are on different vertical scales. Standard errors are clustered on state.

Figure 3. Marginal Effect of the Quality of the Origin Occupation, by Period.



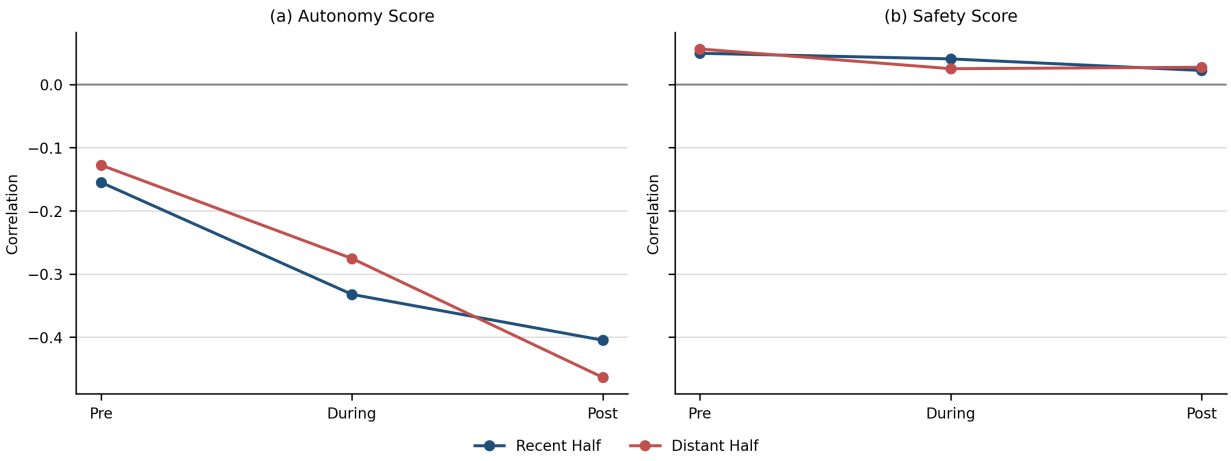
Notes: Marginal effects computed within each period, with ninety-five percent confidence intervals, across the interquartile range of each score (3.43 for autonomy, 2.72 for safety). On the occupation change panel each effect holds the quality of the destinations fixed. Effects are in percentage points against base rates of 90.4 percent for employment at both dates and 29.5 percent for occupation change. Pre is January 2018 through February 2020, during is March 2020 through December 2022, and post is January 2023 through March 2025. The two panels are on different vertical scales. A higher safety score denotes less exposure. Standard errors are clustered on origin occupation. Online Appendix Figure OA11 shows the same estimates with standard errors clustered on state.

Figure 4. Marginal Effect of the Pay of the Origin Occupation, by Period.



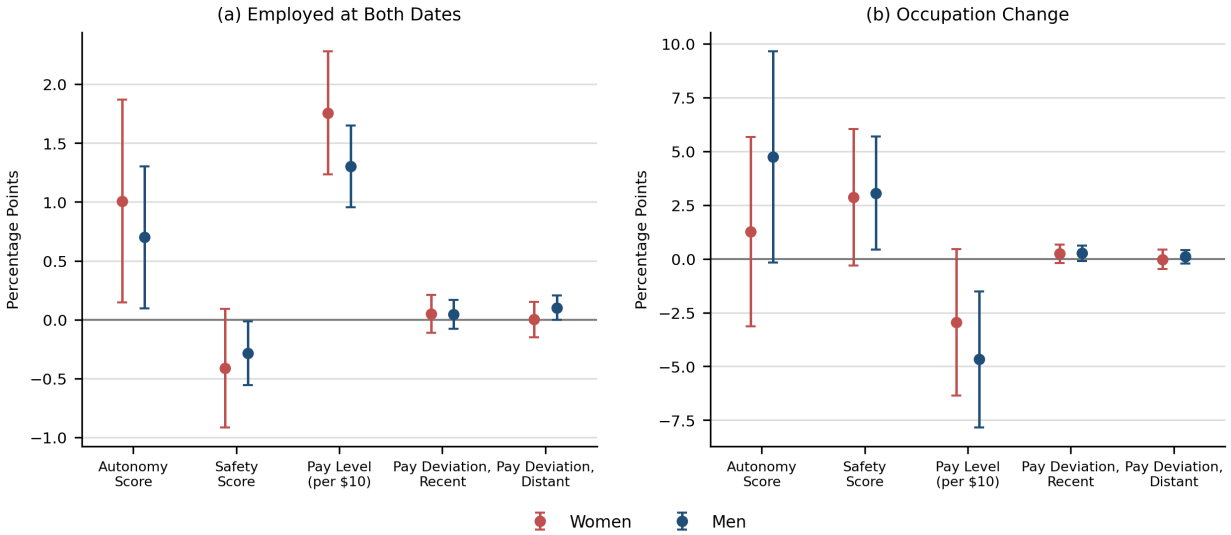
Notes: See notes to Figure 3. Pay level is the 2016-17 mean pay of the origin occupation, in 2017 dollars, per $10 an hour. For each half of the twelve months before the second date, the recent six months and the six months before them, the pay deviation is the proportional difference between the origin occupation's pay in that half and its pay in 2016-17, across its interquartile range (0.092 and 0.085). On the occupation change panel, the effect of occupation pay holds the pay of the destinations fixed. Standard errors are clustered on origin occupation. Online Appendix Figure OA12 shows the same estimates with standard errors clustered on state.

Figure 5. Correlation Between the Occupation Pay Deviation and Each Quality Score, by Period.



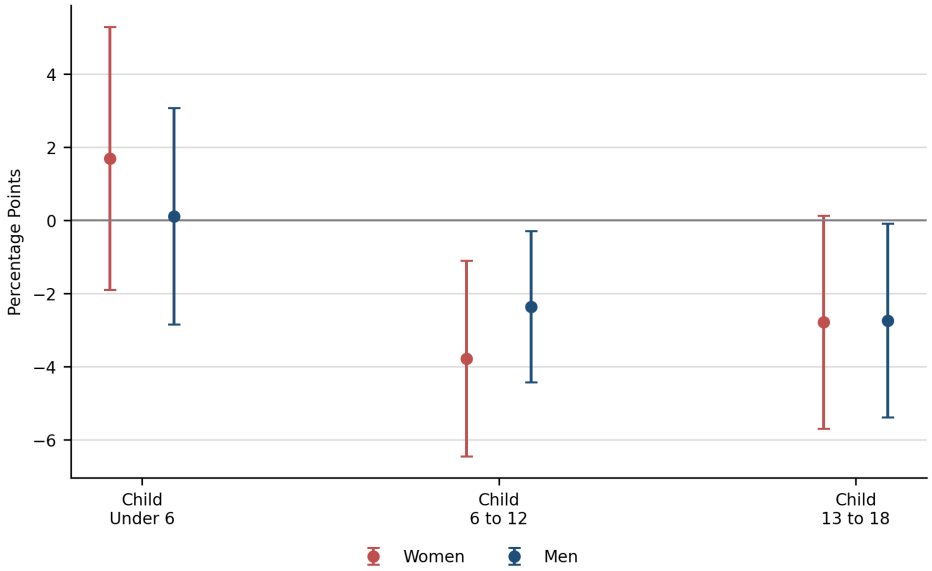
Notes: Correlations across all 1,601,379 records, computed after removing the mean of each measure within each month. For each half of the twelve months before the second date, the recent six months and the six months before them, the pay deviation is the proportional difference between the origin occupation's pay in that half and its pay in 2016-17. Pre is January 2018 through February 2020, during is March 2020 through December 2022, and post is January 2023 through March 2025.

Figure 6. Marginal Effect of the Quality and Pay of the Origin Occupation, by Gender.



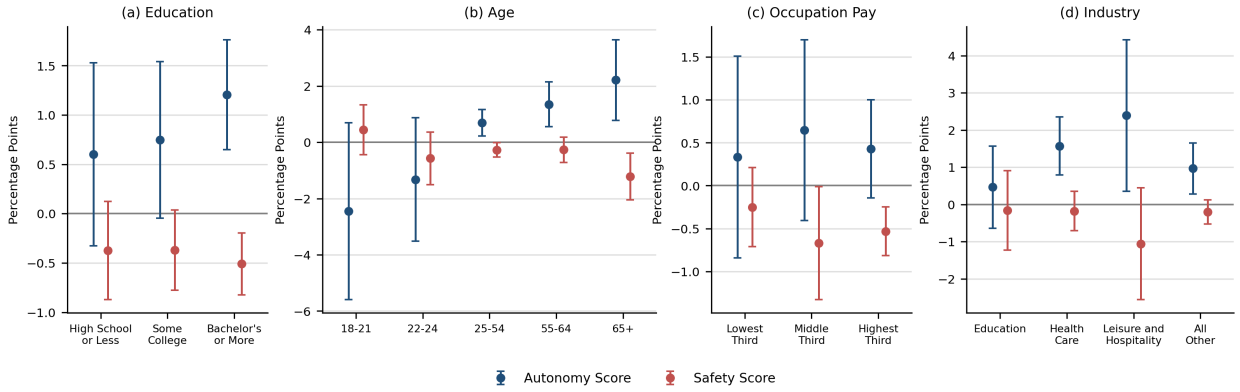
Notes: From the gender model with three child bands, in which gender is interacted with every variable. For each half of the twelve months before the second date, the recent six months and the six months before them, the pay deviation is the proportional difference between the origin occupation's pay in that half and its pay in 2016-17. Scores and pay deviations are across their interquartile ranges (autonomy score 3.43, safety score 2.72, recent and distant pay deviations 0.092 and 0.085) and occupation pay per $10 an hour. On the occupation change panel the effects of the two scores and of occupation pay hold the quality and pay of the destinations fixed. Effects are in percentage points against base rates of 89.2 percent for women and 91.5 percent for men for employment at both dates, and 28.7 and 30.1 percent for occupation change. The two panels are on different vertical scales. Standard errors are clustered on origin occupation. Online Appendix Figure OA13 shows the same estimates with standard errors clustered on state.

Figure 7. Marginal Effect of School Closure on Employment at Both Dates, by Gender and the Presence of a Child in Each Age Band.



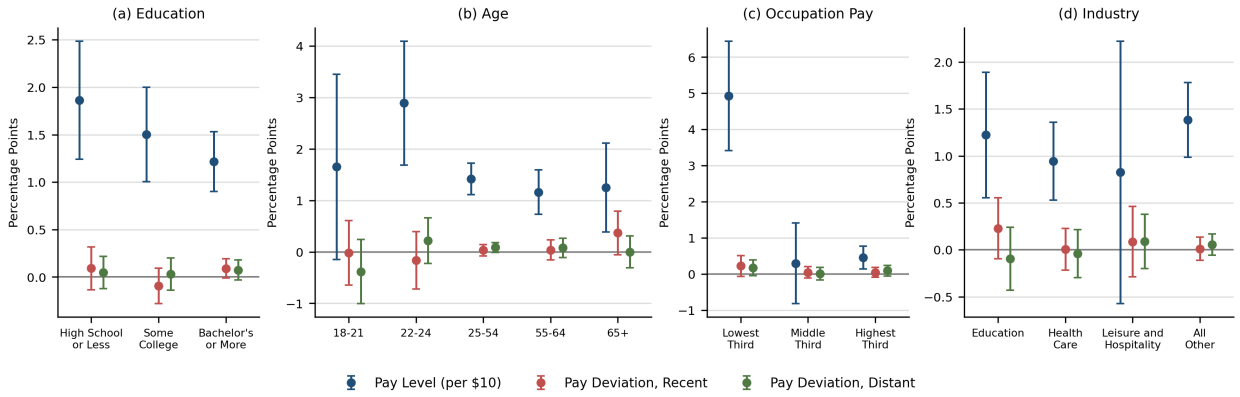
Notes: See notes to Figure 6. Each point is the effect of moving from no local schools closed to all of them closed for the school year, for workers with a child in the band named; the closure rate is defined in the notes to Figure 2. The bands are not mutually exclusive, so a worker with children of more than one age appears in more than one band. Standard errors are clustered on state.

Figure 8. Heterogeneity in the Marginal Effect of the Quality of the Origin Occupation on Employment at Both Dates.



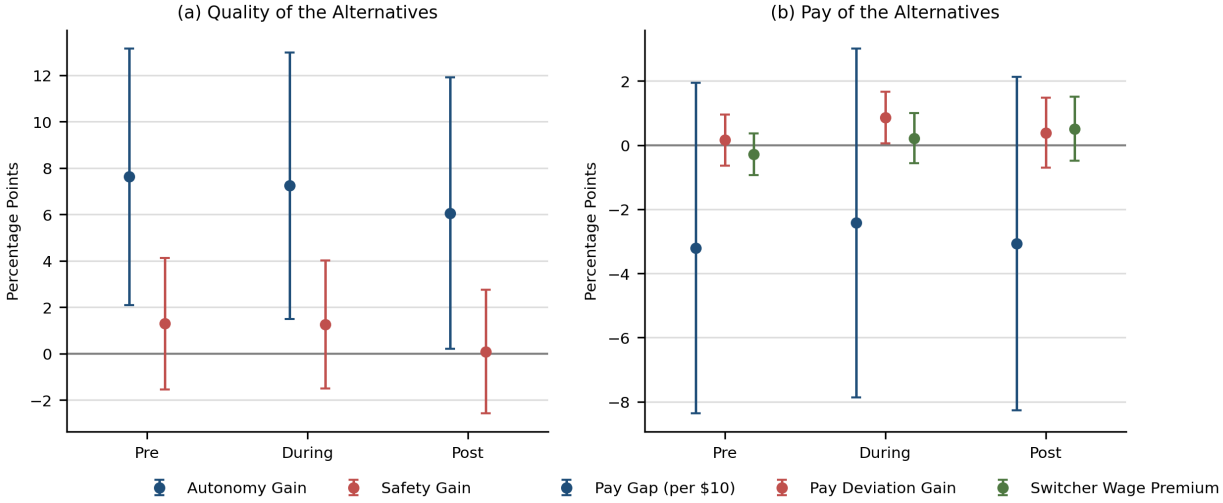
Notes: Each panel is a separate estimation of the main model for the groups named, so no coefficient is constrained across groups. Marginal effects with ninety-five percent confidence intervals, across the interquartile range of each score, in percentage points. Base rates for employment at both dates are 87.8, 89.5 and 93.0 percent by education; 74.9, 85.6, 93.9, 90.3 and 77.0 percent by age; 86.3, 91.3 and 93.9 percent by occupation pay third; and 91.0, 92.1, 84.8 and 90.8 percent for education, health care, leisure and hospitality, and all other industries. Education is high school or less, some college, and a bachelor's degree or more; industry groups are education, health care, leisure and hospitality, and all other industries. Panels are on different vertical scales. Standard errors are clustered on origin occupation within each group. Online Appendix Figure OA14 shows the same estimates with standard errors clustered on state.

Figure 9. Heterogeneity in the Marginal Effect of the Pay of the Origin Occupation on Employment at Both Dates.



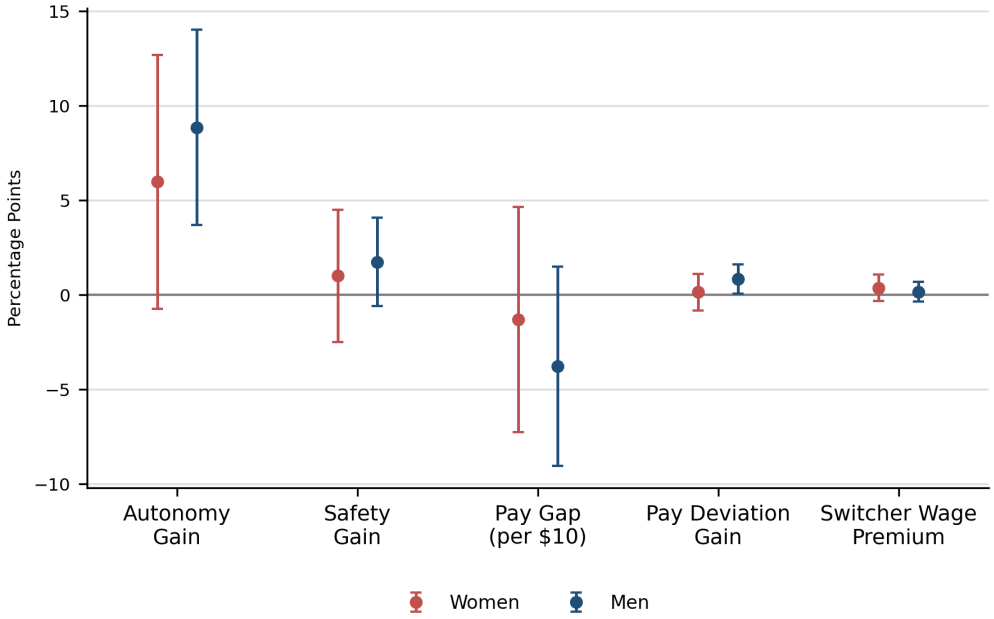
Notes: See notes to Figures 4 and 8. Standard errors are clustered on origin occupation. Online Appendix Figure OA15 shows the same estimates with standard errors clustered on state.

Figure 10. Marginal Effect of the Outside Options on the Probability of an Occupation Change, by Period.



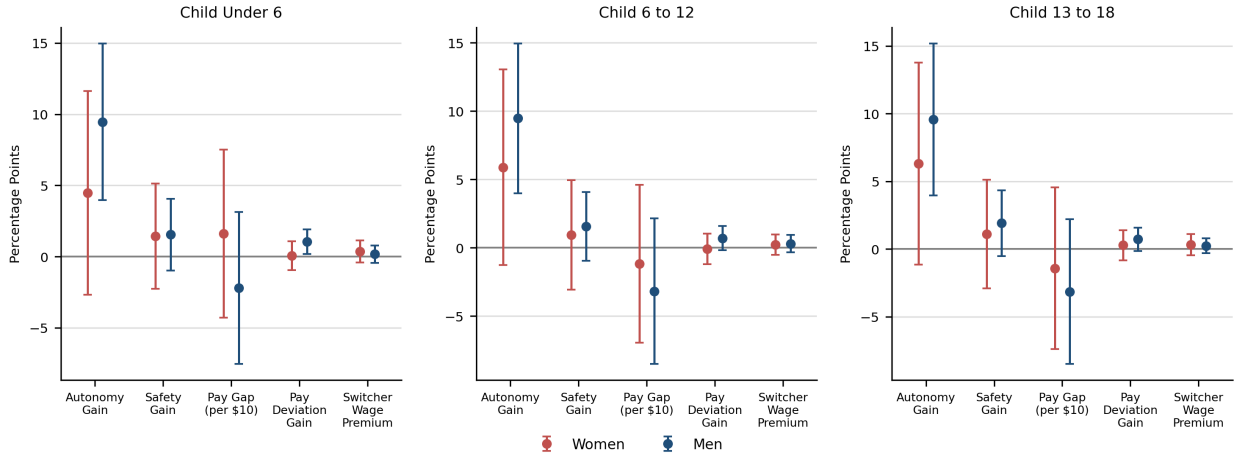
Notes: See notes to Figure 3. These terms enter the occupation equation only. The autonomy gain and safety gain are the average change in each score among workers who left the origin occupation, across the interquartile range of the destination level (1.73 and 1.06), since the effect of the gain is the effect of the destination level. The pay gap is the 2016-17 pay of the destination occupations minus that of the origin occupation, in 2017 dollars, per $10 an hour. The pay deviation gain is how far the destination occupations' pay over the preceding twelve months stood above their 2016-17 level relative to the same measure for the origin occupation, and the switcher wage premium is the individual wage growth of workers who changed occupation minus that of workers who stayed, in the same two-digit origin group and period; both are across their interquartile ranges (0.051 and 0.048). The two panels are on different vertical scales. Standard errors are clustered on origin occupation. Online Appendix Figure OA16 shows the same estimates with standard errors clustered on state.

Figure 11. Marginal Effect of the Outside Options on the Probability of an Occupation Change, by Gender.



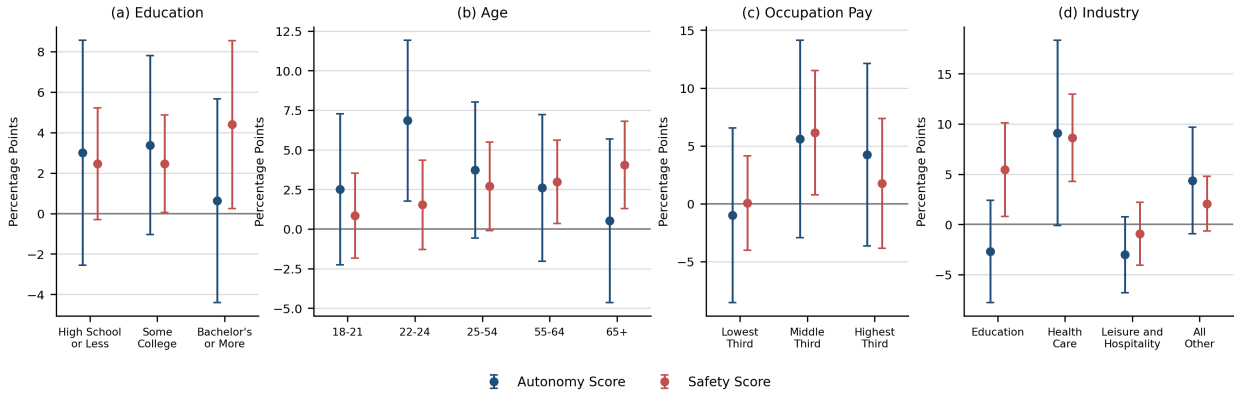
Notes: See notes to Figures 6 and 10. These terms enter the occupation equation only. Standard errors are clustered on origin occupation. Online Appendix Figure OA17 shows the same estimates with standard errors clustered on state.

Figure 12. Marginal Effect of the Outside Options on the Probability of an Occupation Change, by Gender and the Presence of a Child in Each Age Band.



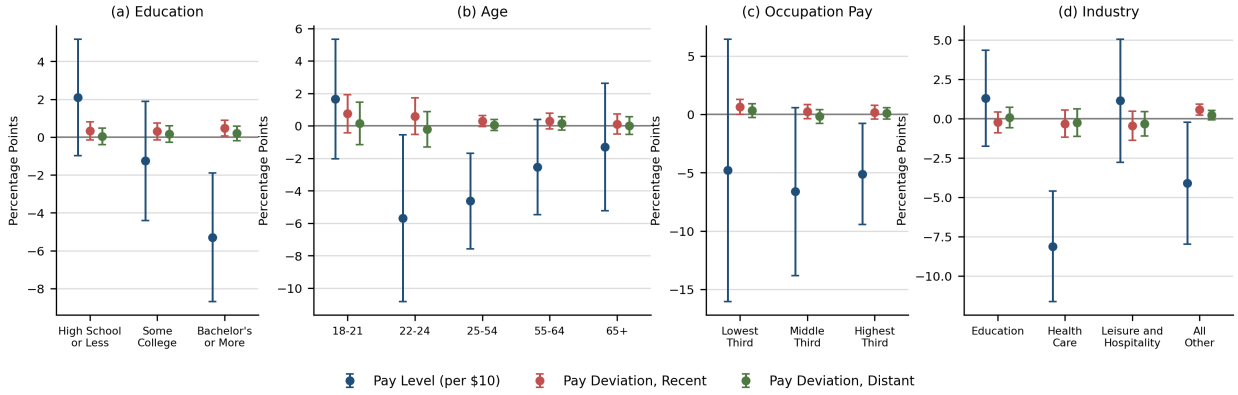
Notes: See notes to Figures 6, 7 and 10. Each point is the effect of that outside-option measure for workers with a child in the band named. Standard errors are clustered on origin occupation. Online Appendix Figure OA18 shows the same estimates with standard errors clustered on state.

Figure 13. Heterogeneity in the Marginal Effect of the Quality of the Origin Occupation on the Probability of an Occupation Change.



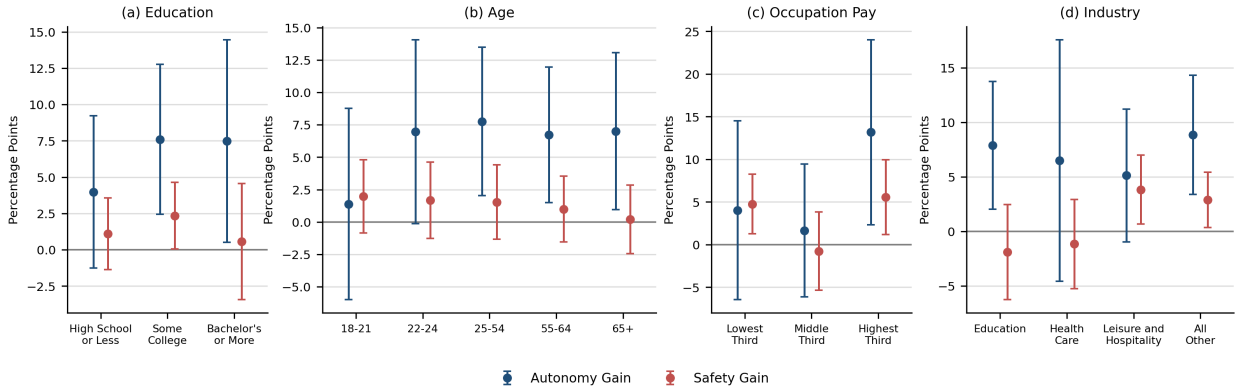
Notes: See notes to Figure 8. Effects are in percentage points and hold the quality of the destinations fixed. Base rates for occupation change are 30.9, 31.2 and 27.3 percent by education; 47.4, 42.8, 29.6, 26.3 and 25.0 percent by age; 32.3, 28.4 and 27.7 percent by occupation pay third; and 17.2, 25.6, 31.2 and 31.6 percent for education, health care, leisure and hospitality, and all other industries. Standard errors are clustered on origin occupation. Online Appendix Figure OA19 shows the same estimates with standard errors clustered on state.

Figure 14. Heterogeneity in the Marginal Effect of the Pay of the Origin Occupation on the Probability of an Occupation Change.



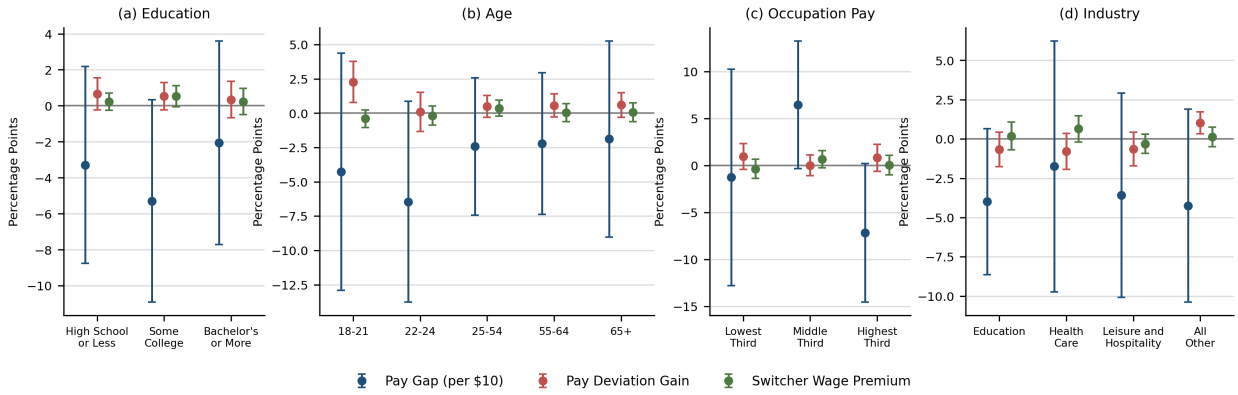
Notes: See notes to Figures 4 and 13. The effect of occupation pay holds the pay of the destinations fixed. Standard errors are clustered on origin occupation. Online Appendix Figure OA20 shows the same estimates with standard errors clustered on state.

Figure 15. Heterogeneity in the Marginal Effect of the Quality of the Outside Options on the Probability of an Occupation Change.



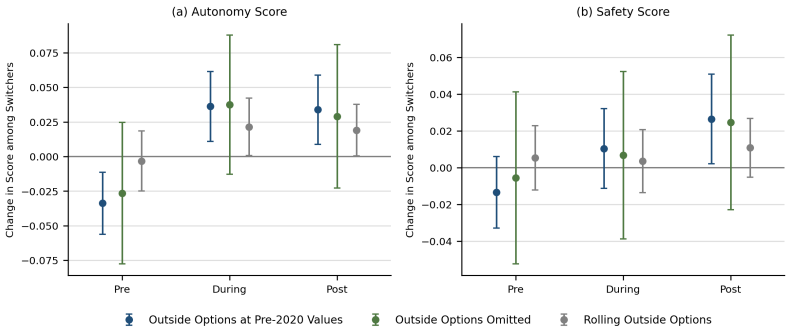
Notes: See notes to Figures 10 and 13. These terms enter the occupation equation only. Standard errors are clustered on origin occupation. Online Appendix Figure OA21 shows the same estimates with standard errors clustered on state.

Figure 16. Heterogeneity in the Marginal Effect of the Pay of the Outside Options on the Probability of an Occupation Change.



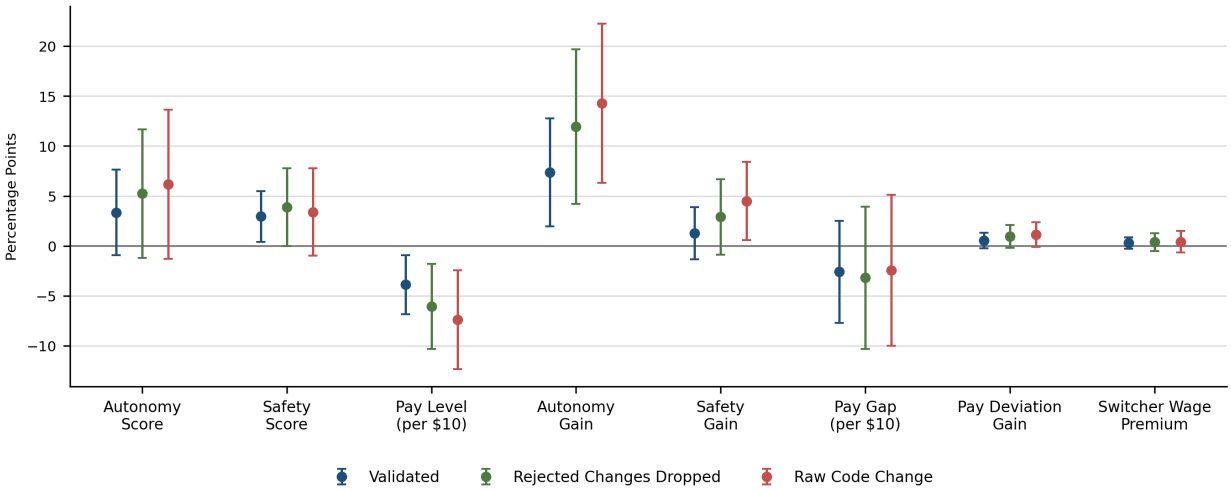
Notes: See notes to Figures 10 and 13. Standard errors are clustered on origin occupation. Online Appendix Figure OA22 shows the same estimates with standard errors clustered on state.

Figure 17. Adjusted Change in Job Quality Among Workers Who Changed Occupation, by Period and Treatment of the Outside Options.



Notes: Adjusted mean change in each score, the occupation held at t minus the occupation held twelve months earlier, among the 426,522 workers with a validated occupation change, from equation 3, with ninety-five percent confidence intervals, under three treatments of the outside options. The estimate used in the text holds them at their pre-2020 values for each origin occupation, and the version that omits them is a check on that construction. The rolling measures used in equation 2 are shown for comparison only, because each is nearly an average of the outcome for other workers leaving the same occupation and so absorbs the shift being measured. Changes are in points of each score; the interquartile ranges are 3.4 for autonomy and 2.7 for safety. Pre is January 2018 through February 2020, during is March 2020 through December 2022, and post is January 2023 through March 2025. Standard errors are clustered on origin occupation. Online Appendix Figure OA23 shows the same estimates with standard errors clustered on state.

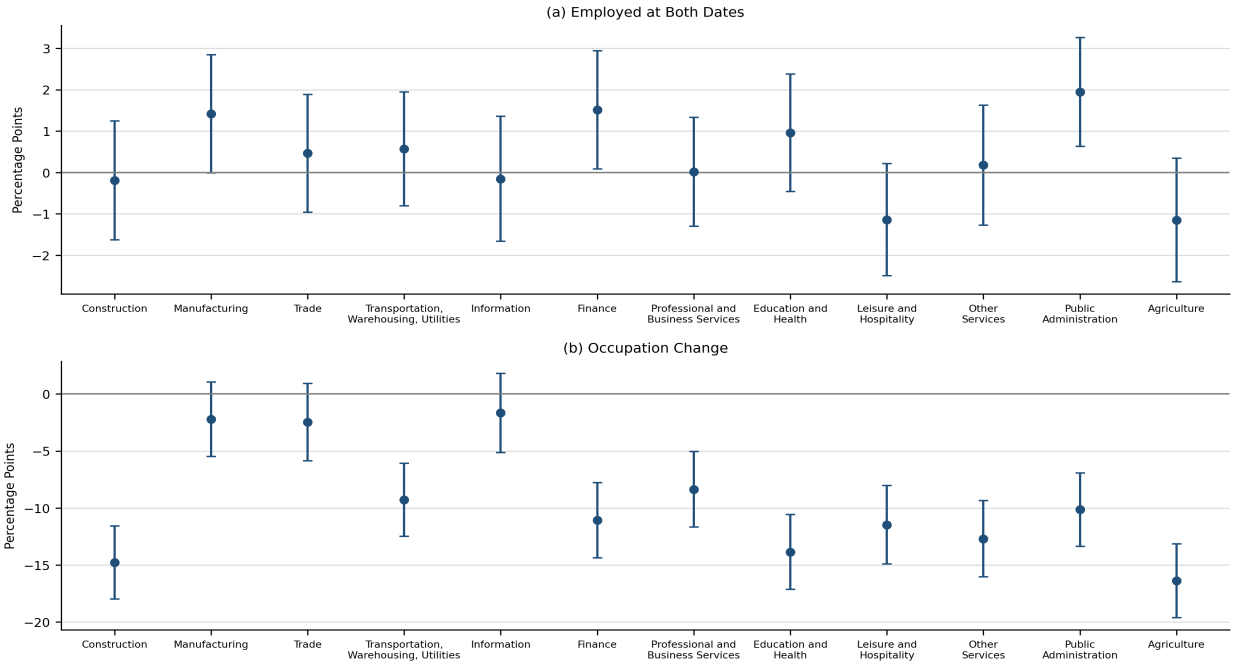
Figure 18. Marginal Effects on Occupation Change Under Three Definitions of the Dependent Variable.



Notes: Marginal effects with ninety-five percent confidence intervals. The effects of the autonomy score, safety score and pay level of the occupation held are shown holding the destinations fixed. See notes to Figures 4 and 10 for units. Validated is the measure used throughout the paper, in which a code change rejected by the screen is coded as a non-change. Rejected changes dropped keeps the records whose code did not change and those whose change passed the screen, and drops the rest. Raw code change applies no screen. Base rates are 29.5, 36.9 and 49.6 percent respectively, so a percentage point does not mean the same thing in each. Standard errors are clustered on origin occupation. Online Appendix Figure OA24 shows the same estimates with standard errors clustered on state.

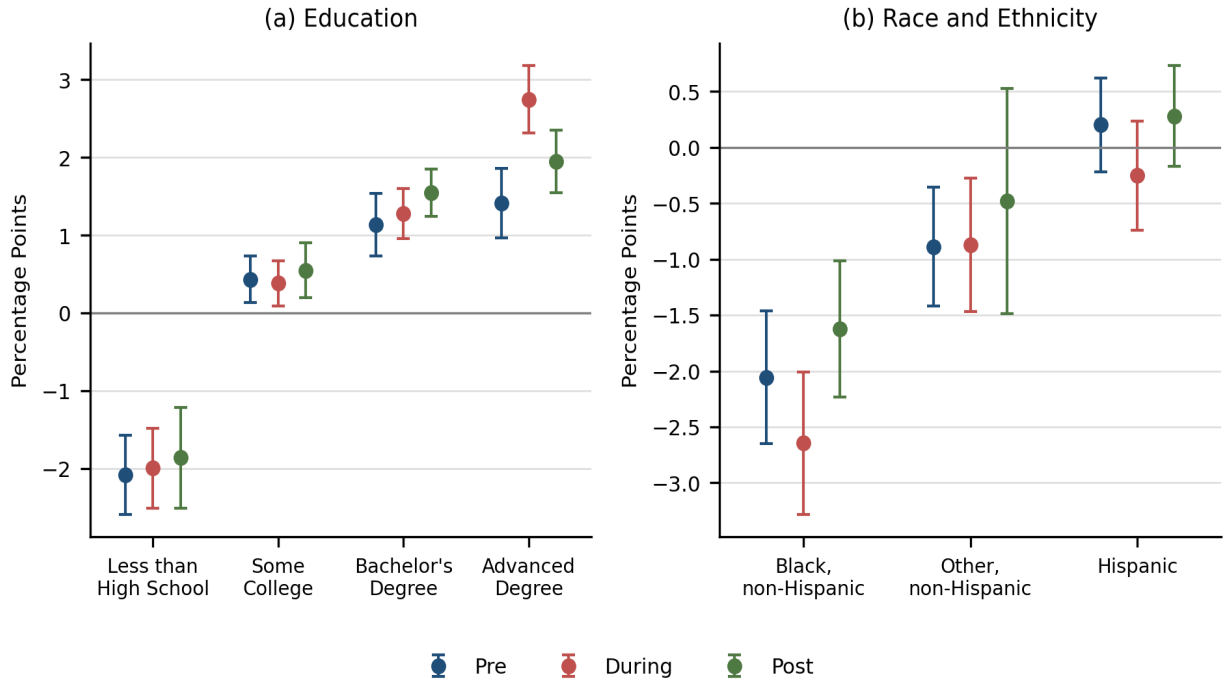
Appendix A. Additional Figures

Figure A1. Marginal Effects of Industry.



Notes: Average marginal effects from the main model, with ninety-five percent confidence intervals, relative to mining, the omitted industry. Industry pay is constant within broad industry, so each effect sets the industry indicator and the industry pay level together at that industry's values. Effects are in percentage points against base rates of 90.4 percent for employment at both dates and 29.5 percent for occupation change. The two panels are on different vertical scales. Standard errors are clustered on state.

Figure A2. Marginal Effects of Education and of Race and Ethnicity on Employment at Both Dates, by Period.



Notes: Average marginal effects on employment at both dates computed within each period, with ninety-five percent confidence intervals, from the model allowing the response to differ across the three periods. Pre is January 2018 through February 2020, during is March 2020 through December 2022, and post is January 2023 through March 2025. Both sets of indicators are interacted with period. See the notes to Table B1 for the omitted categories. The corresponding differences on occupation change do not vary significantly across the periods and are not shown.

Appendix B. Tables of Parameter Estimates and Marginal Effects

Table B1. Main Model.

	Employed at Both Dates	Occupation Change
State Policies and Mandates		
Mask Mandate, Recent Half	-0.0629***	0.0108
	(0.0219)	(0.0203)
Mask Mandate, Distant Half	0.0196	-0.0260
	(0.0227)	(0.0231)
Stay-at-Home Order, Recent Half	-0.0555**	-0.0226
	(0.0264)	(0.0239)
Stay-at-Home Order, Distant Half	0.0823**	-0.0216
	(0.0321)	(0.0278)
Business Closure, Recent Half	-0.0101	0.0452**
	(0.0262)	(0.0214)
Business Closure, Distant Half	-0.0027	-0.0390
	(0.0310)	(0.0281)
Vaccine Mandate, Recent Half	0.0181	0.0165
	(0.0197)	(0.0132)
Vaccine Mandate, Distant Half	-0.0219	-0.0006
	(0.0182)	(0.0151)
Extended Unemployment Benefits, Recent Half	-0.0280	-0.0123
	(0.0234)	(0.0155)
Extended Unemployment Benefits, Distant Half	0.0645***	0.0085
	(0.0233)	(0.0177)
School Closure Rate	-0.1098**	0.0248
	(0.0433)	(0.0364)
School Year with Closure Data	-0.0269	0.0263
	(0.0267)	(0.0186)
Occupational Characteristics		
Autonomy Score	0.0145***	0.1566***
	(0.0017)	(0.0053)
Safety Score	-0.0085***	0.0679***
	(0.0014)	(0.0026)
Occupation Pay	0.0101***	-0.0193***
	(0.0004)	(0.0009)
Occupation Pay Deviation, Recent Half	0.0299	0.0944***
	(0.0234)	(0.0235)

	Employed at Both Dates	Occupation Change
Occupation Pay Deviation, Distant Half	0.0445	0.0294
	(0.0312)	(0.0310)
Industry Pay	0.0039	0.0276***
	(0.0027)	(0.0026)
Industry Pay Deviation	0.0318	0.1273
	(0.1055)	(0.1127)
Own Industry Demand Shock	0.1522**	0.0757
	(0.0591)	(0.0511)
Outside Options		
Autonomy Gain		0.1274***
		(0.0063)
Safety Gain		0.0355***
		(0.0036)
Pay Gap		-0.0077***
		(0.0012)
Pay Deviation Gain		0.3163***
		(0.0571)
Switcher Wage Premium		0.1799***
		(0.0259)
Industry		
Construction	0.0215	-0.1968***
	(0.0299)	(0.0301)
Manufacturing	0.1214***	0.1395***
	(0.0320)	(0.0307)
Trade	0.0886***	0.3483***
	(0.0250)	(0.0259)
Transportation, Warehousing and Utilities	0.0621**	-0.0768**
	(0.0306)	(0.0318)
Information	0.0038	0.0489
	(0.0406)	(0.0409)
Finance	0.1150***	-0.2059***
	(0.0375)	(0.0372)
Professional and Business Services	0.0150	-0.1341***
	(0.0344)	(0.0382)
Education and Health	0.0945***	-0.1707***
	(0.0304)	(0.0296)
Leisure and Hospitality	0.0129	0.2441***
	(0.0298)	(0.0330)

	Employed at Both Dates	Occupation Change
Other Services	0.0627***	-0.0004
	(0.0240)	(0.0280)
Public Administration	0.1344***	-0.2580***
	(0.0396)	(0.0420)
Socioeconomic Characteristics		
Woman	-0.0959***	-0.0042
	(0.0081)	(0.0051)
Age	0.0867***	-0.0291***
	(0.0013)	(0.0013)
Age Squared	-0.0010***	0.0002***
	(0.0000)	(0.0000)
Less than High School	-0.1169***	-0.0455***
	(0.0097)	(0.0112)
Some College	0.0282***	0.0314***
	(0.0061)	(0.0072)
Bachelor's Degree	0.0843***	0.0155**
	(0.0073)	(0.0072)
Advanced Degree	0.1354***	-0.0955***
	(0.0097)	(0.0093)
Black, non-Hispanic	-0.1327***	0.1371***
	(0.0132)	(0.0108)
Other, non-Hispanic	-0.0514***	0.0248*
	(0.0174)	(0.0135)
Hispanic	0.0024	0.0429***
	(0.0098)	(0.0093)
Non-Metropolitan	0.0009	-0.0485***
	(0.0083)	(0.0113)
Metropolitan Status Not Identified	-0.0115	-0.0424
	(0.0275)	(0.0498)
Household and Spousal Characteristics		
Spouse Employed	0.4331***	-0.0101
	(0.0154)	(0.0075)
Married or Partnered	-0.1227***	-0.0087
	(0.0234)	(0.0140)
Spouse Occupation Pay	-0.0052***	-0.0009**
	(0.0006)	(0.0004)
Spouse Occupation Autonomy Score	0.0105***	-0.0003
	(0.0030)	(0.0020)
Spouse Occupation Safety Score	-0.0109***	0.0038***

	Employed at Both Dates	Occupation Change
	(0.0021)	(0.0014)
Spouse Pay Deviation, Recent Half	-0.0183	-0.0370
	(0.0364)	(0.0277)
Spouse Pay Deviation, Distant Half	0.0537	0.0017
	(0.0418)	(0.0326)
Spouse Industry Demand Shock	-0.0037	-0.0646
	(0.0636)	(0.0645)
Spouse Industry Not Determined	-0.0860***	-0.0339***
	(0.0225)	(0.0105)
Spouse Has No Occupation	0.0796***	0.0191
	(0.0194)	(0.0131)
Child Under 6	0.1616***	-0.0322***
	(0.0113)	(0.0080)
Child 6 to 12	0.0843***	-0.0049
	(0.0097)	(0.0058)
Child 13 to 18	0.0221***	0.0438***
	(0.0081)	(0.0065)
Gender Interacted with Children		
Woman × Child Under 6	-0.3625***	-0.0093
	(0.0146)	(0.0104)
Woman × Child 6 to 12	-0.1444***	-0.0090
	(0.0129)	(0.0099)
Woman × Child 13 to 18	-0.0691***	-0.0077
	(0.0123)	(0.0082)

Notes: Probit coefficients, with standard errors clustered on state in parentheses. The marginal effects are in the figures. State and year-month fixed effects are included. The employment equation has 1,601,379 records and the occupation equation the 1,448,086 employed at both dates. Omitted categories are mining, high school graduate, white non-Hispanic and metropolitan. Industry pay is constant within broad industry, so one industry indicator, agriculture, is absorbed and has no coefficient; the marginal effects in Appendix Figure A1 set each industry and its pay together. The child coefficients are for men, and the block that interacts gender with children gives the difference for women. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B2. Model Allowing the Response to Differ Across Pandemic Periods.

	Employed at Both Dates	Occupation Change
State Policies and Mandates		
Mask Mandate, Recent Half	-0.0664***	0.0110
	(0.0218)	(0.0206)
Mask Mandate, Distant Half	0.0162	-0.0263
	(0.0228)	(0.0243)
Stay-at-Home Order, Recent Half	-0.0447*	-0.0264
	(0.0262)	(0.0231)
Stay-at-Home Order, Distant Half	0.0850**	-0.0438
	(0.0339)	(0.0278)
Business Closure, Recent Half	-0.0078	0.0440**
	(0.0261)	(0.0224)
Business Closure, Distant Half	-0.0021	-0.0357
	(0.0320)	(0.0287)
Vaccine Mandate, Recent Half	0.0182	0.0169
	(0.0193)	(0.0123)
Vaccine Mandate, Distant Half	-0.0217	-0.0033
	(0.0182)	(0.0153)
Extended Unemployment Benefits, Recent Half	-0.0290	-0.0086
	(0.0239)	(0.0154)
Extended Unemployment Benefits, Distant Half	0.0675***	0.0121
	(0.0239)	(0.0183)
School Closure Rate	-0.1121**	0.0380
	(0.0452)	(0.0345)
School Year with Closure Data	-0.0510	-0.0409
	(0.0417)	(0.0429)
Occupational Characteristics		
Autonomy Score	0.0205***	0.1280***
	(0.0040)	(0.0099)
Safety Score	-0.0378***	0.0428***
	(0.0076)	(0.0115)
Occupation Pay	0.0121***	-0.0230***
	(0.0007)	(0.0012)
Occupation Pay Deviation, Recent Half	-0.0670	0.1107**
	(0.0519)	(0.0487)
Occupation Pay Deviation, Distant Half	-0.1068*	-0.0649

	Employed at Both Dates	Occupation Change
	(0.0622)	(0.0483)
Industry Pay	0.0026	0.0260***
	(0.0025)	(0.0026)
Industry Pay Deviation	-0.0673	0.0654
	(0.1166)	(0.1202)
Own Industry Demand Shock	0.1309**	0.0748
	(0.0587)	(0.0517)
Outside Options		
Autonomy Gain		0.1329***
		(0.0100)
Safety Gain		0.0368***
		(0.0065)
Pay Gap		-0.0097***
		(0.0016)
Pay Deviation Gain		0.0946
		(0.0895)
Switcher Wage Premium		-0.1789***
		(0.0542)
Industry		
Construction	0.0151	-0.1971***
	(0.0288)	(0.0298)
Manufacturing	0.1232***	0.1346***
	(0.0313)	(0.0307)
Trade	0.0973***	0.3241***
	(0.0240)	(0.0256)
Transportation, Warehousing and Utilities	0.0705**	-0.0813**
	(0.0296)	(0.0317)
Information	0.0221	0.0505
	(0.0388)	(0.0406)
Finance	0.1365***	-0.2039***
	(0.0358)	(0.0371)
Professional and Business Services	0.0421	-0.1382***
	(0.0329)	(0.0378)
Education and Health	0.1048***	-0.1713***
	(0.0292)	(0.0293)
Leisure and Hospitality	0.0328	0.1956***
	(0.0293)	(0.0328)
Other Services	0.0707***	-0.0059
	(0.0235)	(0.0278)

	Employed at Both Dates	Occupation Change
Public Administration	0.1571***	-0.2461***
	(0.0378)	(0.0421)
Socioeconomic Characteristics		
Woman	-0.0942***	-0.0059
	(0.0081)	(0.0051)
Age	0.0862***	-0.0281***
	(0.0013)	(0.0013)
Age Squared	-0.0010***	0.0002***
	(0.0000)	(0.0000)
Less than High School	-0.1268***	-0.0474***
	(0.0151)	(0.0114)
Some College	0.0291***	0.0367***
	(0.0103)	(0.0107)
Bachelor's Degree	0.0788***	0.0211*
	(0.0144)	(0.0111)
Advanced Degree	0.0992***	-0.1079***
	(0.0166)	(0.0153)
Black, non-Hispanic	-0.1327***	0.1467***
	(0.0185)	(0.0158)
Other, non-Hispanic	-0.0598***	0.0159
	(0.0177)	(0.0152)
Hispanic	0.0143	0.0500***
	(0.0153)	(0.0120)
Non-Metropolitan	-0.0008	-0.0472***
	(0.0083)	(0.0113)
Metropolitan Status Not Identified	-0.0125	-0.0411
	(0.0281)	(0.0499)
Household and Spousal Characteristics		
Spouse Employed	0.4314***	-0.0087
	(0.0155)	(0.0074)
Married or Partnered	-0.1250***	-0.0078
	(0.0231)	(0.0141)
Spouse Occupation Pay	-0.0051***	-0.0009**
	(0.0006)	(0.0004)
Spouse Occupation Autonomy Score	0.0102***	-0.0005
	(0.0030)	(0.0020)
Spouse Occupation Safety Score	-0.0104***	0.0032**
	(0.0021)	(0.0014)
Spouse Pay Deviation, Recent Half	-0.0213	-0.0356

	Employed at Both Dates	Occupation Change
	(0.0362)	(0.0278)
Spouse Pay Deviation, Distant Half	0.0488	0.0022
	(0.0415)	(0.0332)
Spouse Industry Demand Shock	0.0349	-0.0693
	(0.0629)	(0.0646)
Spouse Industry Not Determined	-0.0867***	-0.0339***
	(0.0224)	(0.0104)
Spouse Has No Occupation	0.0826***	0.0180
	(0.0192)	(0.0132)
Child Under 6	0.1610***	-0.0315***
	(0.0112)	(0.0081)
Child 6 to 12	0.0836***	-0.0053
	(0.0097)	(0.0059)
Child 13 to 18	0.0229***	0.0429***
	(0.0081)	(0.0064)
Occupation Quality Interacted with Period		
During × Autonomy Score	0.0329***	-0.0191*
	(0.0046)	(0.0115)
Post × Autonomy Score	0.0113**	-0.0288***
	(0.0051)	(0.0107)
During × Safety Score	0.0084**	-0.0026
	(0.0043)	(0.0075)
Post × Safety Score	-0.0011	-0.0275***
	(0.0041)	(0.0069)
Occupation Pay Interacted with Period		
During × Occupation Pay	0.0007	0.0025*
	(0.0010)	(0.0015)
Post × Occupation Pay	-0.0015	0.0005
	(0.0010)	(0.0018)
During × Occupation Pay Deviation, Recent Half	0.1896***	-0.0013
	(0.0660)	(0.0843)
Post × Occupation Pay Deviation, Recent Half	0.1419**	-0.1937**
	(0.0723)	(0.0760)
During × Occupation Pay Deviation, Distant Half	0.2711***	0.1224*
	(0.0689)	(0.0710)
Post × Occupation Pay Deviation, Distant Half	0.1172	-0.0366

	Employed at Both Dates	Occupation Change
	(0.0849)	(0.0783)
Outside Options Interacted with Period		
During × Autonomy Gain		-0.0083
		(0.0147)
Post × Autonomy Gain		-0.0286**
		(0.0127)
During × Safety Gain		-0.0015
		(0.0085)
Post × Safety Gain		-0.0343***
		(0.0079)
During × Pay Gap		0.0024
		(0.0020)
Post × Pay Gap		0.0005
		(0.0023)
During × Pay Deviation Gain		0.4107***
		(0.1412)
Post × Pay Deviation Gain		0.1334
		(0.1421)
During × Switcher Wage Premium		0.3099***
		(0.0818)
Post × Switcher Wage Premium		0.4935***
		(0.1017)
Education Interacted with Period		
During × Less than High School	0.0202	0.0279
	(0.0218)	(0.0190)
Post × Less than High School	0.0151	-0.0256
	(0.0196)	(0.0188)
During × Some College	-0.0071	0.0031
	(0.0129)	(0.0116)
Post × Some College	0.0069	-0.0206
	(0.0167)	(0.0138)
During × Bachelor's Degree	-0.0033	-0.0040
	(0.0158)	(0.0130)
Post × Bachelor's Degree	0.0273	-0.0137
	(0.0175)	(0.0139)
During × Advanced Degree	0.0721***	0.0248
	(0.0204)	(0.0177)
Post × Advanced Degree	0.0371*	0.0153
	(0.0207)	(0.0169)

	Employed at Both Dates	Occupation Change
Race and Ethnicity Interacted with Period		
During × Black, non-Hispanic	-0.0157	-0.0251
	(0.0192)	(0.0154)
Post × Black, non-Hispanic	0.0275	-0.0132
	(0.0243)	(0.0205)
During × Other, non-Hispanic	0.0083	0.0093
	(0.0236)	(0.0176)
Post × Other, non-Hispanic	0.0272	0.0008
	(0.0261)	(0.0175)
During × Hispanic	-0.0296	-0.0128
	(0.0195)	(0.0165)
Post × Hispanic	0.0052	-0.0082
	(0.0252)	(0.0218)
Occupation Quality Interacted with Occupation Pay		
Autonomy Score × Occupation Pay	-0.0010***	0.0015***
	(0.0002)	(0.0002)
Autonomy Score × Occupation Pay Deviation, Recent Half	0.0090	-0.0095
	(0.0163)	(0.0147)
Autonomy Score × Occupation Pay Deviation, Distant Half	-0.0038	0.0367***
	(0.0180)	(0.0123)
Safety Score × Occupation Pay	0.0008***	0.0011***
	(0.0003)	(0.0003)
Safety Score × Occupation Pay Deviation, Recent Half	-0.0074	-0.0452***
	(0.0150)	(0.0123)
Safety Score × Occupation Pay Deviation, Distant Half	0.0086	-0.0146
	(0.0179)	(0.0157)
Autonomy Interacted with Safety		
Autonomy Score × Safety Score	-0.0090***	-0.0011
	(0.0015)	(0.0014)
Occupation Quality Interacted with Policy		
Autonomy Score × Mask Mandate, Recent Half	0.0035	-0.0015
	(0.0076)	(0.0088)
Autonomy Score × Mask Mandate, Distant Half	0.0030	0.0010
	(0.0073)	(0.0079)

	Employed at Both Dates	Occupation Change
Autonomy Score × Stay-at-Home Order, Recent Half	0.0360***	0.0014
	(0.0101)	(0.0109)
Autonomy Score × Stay-at-Home Order, Distant Half	-0.0317**	0.0427***
	(0.0147)	(0.0080)
Autonomy Score × Business Closure, Recent Half	0.0067	0.0043
	(0.0089)	(0.0096)
Autonomy Score × Business Closure, Distant Half	0.0010	-0.0096
	(0.0091)	(0.0089)
Autonomy Score × Vaccine Mandate, Recent Half	0.0019	-0.0021
	(0.0071)	(0.0064)
Autonomy Score × Vaccine Mandate, Distant Half	-0.0134*	0.0031
	(0.0081)	(0.0044)
Autonomy Score × Extended Unemployment Benefits, Recent Half	0.0111*	-0.0056
	(0.0060)	(0.0071)
Autonomy Score × Extended Unemployment Benefits, Distant Half	-0.0096	-0.0098
	(0.0092)	(0.0073)
Autonomy Score × School Closure Rate	0.0332**	-0.0121
	(0.0135)	(0.0136)
Autonomy Score × School Year with Closure Data	-0.0254***	0.0031
	(0.0049)	(0.0046)
Safety Score × Mask Mandate, Recent Half	0.0063	-0.0071
	(0.0085)	(0.0074)
Safety Score × Mask Mandate, Distant Half	-0.0019	-0.0005
	(0.0089)	(0.0080)
Safety Score × Stay-at-Home Order, Recent Half	0.0086	-0.0024
	(0.0106)	(0.0161)

	Employed at Both Dates	Occupation Change
Safety Score × Stay-at-Home Order, Distant Half	-0.0062	0.0057
	(0.0116)	(0.0153)
Safety Score × Business Closure, Recent Half	0.0075	-0.0056
	(0.0095)	(0.0093)
Safety Score × Business Closure, Distant Half	-0.0211**	-0.0139
	(0.0100)	(0.0111)
Safety Score × Vaccine Mandate, Recent Half	0.0021	-0.0059
	(0.0096)	(0.0070)
Safety Score × Vaccine Mandate, Distant Half	0.0003	-0.0020
	(0.0080)	(0.0051)
Safety Score × Extended Unemployment Benefits, Recent Half	-0.0071	0.0154**
	(0.0078)	(0.0072)
Safety Score × Extended Unemployment Benefits, Distant Half	-0.0009	-0.0002
	(0.0069)	(0.0084)
Safety Score × School Closure Rate	0.0316**	-0.0051
	(0.0159)	(0.0138)
Safety Score × School Year with Closure Data	-0.0056	-0.0022
	(0.0051)	(0.0045)
Gender Interacted with Children		
Woman × Child Under 6	-0.3626***	-0.0084
	(0.0146)	(0.0103)
Woman × Child 6 to 12	-0.1442***	-0.0084
	(0.0128)	(0.0098)
Woman × Child 13 to 18	-0.0682***	-0.0087
	(0.0123)	(0.0082)

Notes: See notes to Table B1. This model adds the interaction blocks shown. Period is pre, January 2018 through February 2020; during, March 2020 through December 2022; and post, January 2023 through March 2025, with pre the omitted category. The marginal effects of the period-interacted terms in each period are in Figures 3, 4 and 10 and Appendix Figure A2, and the quality effects at levels of pay and of each policy in Tables B4 and B5.

Table B3. Marginal Effects of Children by Gender, Gender Model.

	Women	Men	Women Minus Men
Employment at Both Dates			
Child Under 6	-3.297***	1.623***	-4.920***
	(0.197)	(0.136)	(0.213)
Child 6 to 12	-0.683***	0.806***	-1.488***
	(0.204)	(0.132)	(0.231)
Child 13 to 18	-0.544***	0.136	-0.680***
	(0.140)	(0.113)	(0.180)
Occupation Change			
Child Under 6	-1.355***	-0.645**	-0.710*
	(0.301)	(0.271)	(0.375)
Child 6 to 12	-0.555*	0.278	-0.833**
	(0.307)	(0.205)	(0.367)
Child 13 to 18	1.212***	1.696***	-0.484*
	(0.248)	(0.220)	(0.288)

Notes: Average marginal effects in percentage points of the presence of a child in each age band, from the gender model with three child bands, in which gender is interacted with every variable, with standard errors in parentheses, clustered on state. The bands are not mutually exclusive. Base rates are 89.2 percent for women and 91.5 percent for men for employment at both dates, and 28.7 and 30.1 percent for occupation change. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B4. Marginal Effects of the Quality Scores at Levels of Occupation Pay, Period Model.

	25th Percentile	Median	75th Percentile	75th Minus 25th
Employment at Both Dates				
Autonomy Score	0.248***	0.141***	0.000	-0.248***
	(0.030)	(0.030)	(0.038)	(0.034)
	[0.089]***	[0.076]*	[0.082]	[0.088]***
Safety Score	-0.321***	-0.219***	-0.081**	0.240***
	(0.048)	(0.027)	(0.034)	(0.065)
	[0.099]***	[0.063]***	[0.058]	[0.112]**
Occupation Change				
Autonomy Score	0.771***	1.086***	1.361***	0.591***
	(0.114)	(0.083)	(0.078)	(0.104)
	[0.715]	[0.640]*	[0.667]**	[0.563]
Safety Score	0.955***	1.164***	1.331***	0.376**
	(0.098)	(0.057)	(0.092)	(0.152)
	[0.721]	[0.561]**	[0.648]**	[0.811]

Notes: Average marginal effects in percentage points per point of each score from the model of Table B2, with occupation pay set at the 25th percentile, the median and the 75th percentile of its distribution in the estimation sample, and the difference between the 75th and 25th percentiles, with standard errors clustered on state in parentheses and clustered on origin occupation in brackets. Stars on the estimates denote significance under state clustering and stars beside the bracketed standard errors significance under occupation clustering. On occupation change, the effects hold the quality of the destinations fixed. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table B5. Marginal Effects of the Quality Scores by Policy, Period Model.

	Not in Force	In Force for the Full Year	Difference
Employment at Both Dates: Mask Mandate			
Autonomy Score	0.350***	0.483***	0.133
	(0.062) [0.123]***	(0.158) [0.170]***	(0.178) [0.145]
Safety Score	-0.151***	-0.081	0.070
	(0.051) [0.086]*	(0.142) [0.148]	(0.173) [0.175]
Employment at Both Dates: Stay-at-Home Order			
Autonomy Score	0.365***	0.421*	0.056
	(0.055) [0.123]***	(0.252) [0.337]	(0.251) [0.337]
Safety Score	-0.140***	-0.092	0.048
	(0.047) [0.077]*	(0.257) [0.357]	(0.284) [0.372]
Employment at Both Dates: Business Closure			
Autonomy Score	0.355***	0.481***	0.127
	(0.071) [0.129]***	(0.156) [0.236]**	(0.191) [0.243]
Safety Score	-0.101*	-0.327	-0.226
	(0.052) [0.088]	(0.201) [0.196]*	(0.234) [0.229]
Employment at Both Dates: Vaccine Mandate			
Autonomy Score	0.392***	0.199	-0.193
	(0.054) [0.125]***	(0.144) [0.150]	(0.135) [0.123]
Safety Score	-0.137***	-0.101	0.035
	(0.035) [0.077]*	(0.160) [0.145]	(0.156) [0.150]
Employment at Both Dates: Extended Unemployment Benefits			
Autonomy Score	0.364***	0.371***	0.008
	(0.078) [0.128]***	(0.128) [0.154]**	(0.173) [0.140]
Safety Score	-0.095	-0.218*	-0.124
	(0.067) [0.078]	(0.113) [0.144]	(0.163) [0.158]
Employment at Both Dates: School Closure Rate			
Autonomy Score	0.123	0.737***	0.614**
	(0.078) [0.112]	(0.204) [0.224]***	(0.249) [0.234]***
Safety Score	-0.275***	0.250	0.525*
	(0.071) [0.094]***	(0.217) [0.192]	(0.274) [0.248]**
Occupation Change: Mask Mandate			

	Not in Force	In Force for the Full Year	Difference
Autonomy Score	0.873***	0.848***	-0.025
	(0.162) [0.723]	(0.322) [0.764]	(0.409) [0.318]
Safety Score	1.079***	0.816***	-0.263
	(0.112) [0.579]*	(0.263) [0.659]	(0.316) [0.357]
Occupation Change: Stay-at-Home Order			
Autonomy Score	0.770***	2.206***	1.436***
	(0.125) [0.724]	(0.438) [0.860]**	(0.464) [0.558]**
Safety Score	1.008***	1.097**	0.090
	(0.092) [0.580]*	(0.482) [0.838]	(0.501) [0.627]
Occupation Change: Business Closure			
Autonomy Score	0.895***	0.724*	-0.171
	(0.147) [0.713]	(0.407) [0.837]	(0.477) [0.409]
Safety Score	1.131***	0.481	-0.650
	(0.125) [0.570]**	(0.367) [0.750]	(0.444) [0.466]
Occupation Change: Vaccine Mandate			
Autonomy Score	0.868***	0.910***	0.043
	(0.126) [0.723]	(0.230) [0.723]	(0.242) [0.265]
Safety Score	1.041***	0.785***	-0.256
	(0.091) [0.582]*	(0.192) [0.609]	(0.190) [0.258]
Occupation Change: Extended Unemployment Benefits			
Autonomy Score	1.016***	0.490**	-0.526*
	(0.140) [0.717]	(0.241) [0.765]	(0.287) [0.278]*
Safety Score	0.864***	1.363***	0.499
	(0.147) [0.594]	(0.265) [0.618]**	(0.369) [0.349]
Occupation Change: School Closure Rate			
Autonomy Score	0.933***	0.533	-0.400
	(0.154) [0.779]	(0.397) [0.710]	(0.467) [0.519]
Safety Score	1.012***	0.851**	-0.160
	(0.160) [0.614]*	(0.362) [0.679]	(0.474) [0.553]

Notes: Average marginal effects in percentage points per point of each score from the model of Table B2, with the policy out of force in both six-month halves and in force for the whole of both halves, and the difference, over records in the pandemic period, March 2020 through December 2022, with standard errors clustered on state in parentheses and clustered on origin occupation in brackets. Stars on the estimates denote significance under state clustering and stars beside the bracketed standard errors significance under occupation clustering. For school closure the comparison is between no local schools closed and all of them closed for the school year, over the 2020-21 and 2021-22 school years. On occupation change, the effects hold the quality of the destinations fixed. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Appendix C. The Joint Model

This appendix sets out why the two equations are estimated separately. A selection model that corrects the occupation-change estimates for selection into employment at both dates needs a variable that shifts employment at both dates without entering the decision to change occupation. We screened five candidates, all measured from the spouse's record: the spouse's industry demand shock, the pay of the spouse's occupation, the spouse's employment, and the recent and distant pay deviations of the spouse's occupation. Each was tested separately for women and for men against four conditions: that it varies in all three periods; that it predicts employment at both dates pooled across the periods at the 5 percent level; that it does so within each period; and that it does not predict occupation change at the 10 percent level, pooled or within any period. No candidate meets all four for either gender (Table C1). The pay of the spouse's occupation, for example, predicts women's employment at both dates strongly in every period but also predicts their occupation change ($p = 0.029$), and the spouse's industry demand shock predicts employment at both dates pooled but not within every period.

Without an exclusion restriction, the correlation between the two equations in a joint model is identified by the assumption of bivariate normality. Table C2 reports a probit with sample selection estimated on the specification of the main model under three choices: no exclusion, the pay of the spouse's occupation excluded from the occupation-change equation, and the spouse's industry demand shock excluded. The inverse hyperbolic tangent of the estimated correlation is -0.439, -0.448 and -0.441, each with a standard error of about 0.07, so the choice of exclusion moves it by less than 0.01, which is what identification from functional form rather than from an exclusion implies.

The joint model's occupation-change estimates are nonetheless close to those of the probit. With no exclusion, the autonomy score raises the probability of changing occupation by 17.87 percentage points across its interquartile range and the safety score by 6.12 across its own, against 17.94 and 6.17 in the main model, and the autonomy gain available elsewhere raises it by 7.34 across its own interquartile range against 7.37. Correcting for selection on unobservables does not change the results we report, and the paper therefore reports the separately estimated probits.

Table C1. Screen of Candidate Exclusion Restrictions: P Values by Gender and Period.

Candidate and Gender	Predicts Employed at Both Dates				Predicts Occupation Change				
	Pooled	**Pre**	**During**	**Post**	**Pooled**	**Pre**	**During**	**Post**	**Meets All Four**
Spouse Industry Demand Shock									
Women	<0.001	0.006	<0.001	0.118	0.451	0.055	0.629	0.803	No
Men	<0.001	0.289	0.070	0.198	0.328	0.020	0.590	0.042	No
Spouse Occupation Pay									
Women	<0.001	<0.001	<0.001	<0.001	0.029	0.061	0.122	0.393	No
Men	<0.001	<0.001	0.058	0.198	0.889	0.829	0.286	0.300	No
Spouse Employed									
Women	<0.001	<0.001	<0.001	<0.001	0.046	0.344	0.246	0.390	No
Men	<0.001	<0.001	<0.001	<0.001	0.081	0.052	0.014	0.055	No
Spouse Occupation Pay Deviation, Recent Half									
Women	0.374	0.055	0.089	0.398	0.298	0.838	0.977	0.627	No
Men	0.915	0.249	0.372	0.364	0.014	0.174	0.213	0.469	No
Spouse Occupation Pay Deviation, Distant Half									
Women	0.047	0.018	0.377	0.312	0.836	0.022	0.830	0.050	No
Men	0.014	0.207	0.342	0.350	0.619	0.054	0.497	0.268	No

Notes: p values of the joint test that the candidate's terms are zero, from the gender models, with standard errors clustered on state. A candidate meets all four conditions if it varies in all three periods, predicts employment at both dates at the 5 percent level pooled and within every period, and does not predict occupation change at the 10 percent level pooled or within any period. Every candidate varies in all three periods. Pre is January 2018 through February 2020, during is March 2020 through December 2022, and post is January 2023 onward.

Table C2. Joint Model of Employment at Both Dates and Occupation Change.

Exclusion from the Occupation Equation	Inverse Hyperbolic Tangent of the Correlation	Standard Error	Correlation	Wald Test	p Value
Correlation between the Two Equations					
No Exclusion	-0.439	(0.068)	-0.413	41.4	<0.001
Spouse Occupation Pay Excluded	-0.448	(0.066)	-0.420	46.2	<0.001
Spouse Industry Demand Shock Excluded	-0.441	(0.066)	-0.414	44.1	<0.001

	Joint Model, No Exclusion	Main Model Probit
Marginal Effects on Occupation Change		
Autonomy Score	5.21	5.23
Safety Score	2.25	2.27
Autonomy Gain	4.24	4.26
Safety Gain	1.17	1.19
Pay Gap (per $10)	-2.56	-2.59
Pay Deviation Gain	10.49	10.57
Switcher Wage Premium	6.33	6.01

Notes: Probit with sample selection, employment at both dates as the selection equation and occupation change among those employed at both dates as the outcome, estimated on the specification of the main model with standard errors clustered on state; N = 1,601,379, of whom 1,448,086 are employed at both dates. The correlation is between the unobserved determinants of the two equations, estimated as its inverse hyperbolic tangent. The Wald test is of a zero correlation, on one degree of freedom. Marginal effects are on the probability of an occupation change among those employed at both dates, in percentage points per point of each score and gain, per $10 an hour of the pay gap, and per unit of the pay deviation gain and switcher wage premium.